

## NCERT Exemplar Solutions for Class 12

### Mathematics

### Chapter 13 – Probability

#### Solved Examples

#### Short Answer Questions

**1. A and B are two candidates seeking admission in a college. The probability that A is selected is 0.7 and the probability that exactly one of them is selected is 0.6. Find the probability that B is selected.**

**Ans:** Let  $p$  be the probability that B gets selected.

$$P(A \text{ is selected}) = 0.7$$

$$\text{Then, } P(A') = 1 - P(A) = 1 - 0.7 = 0.3$$

$$\text{And } P(\text{Exactly one of A, B is selected}) = 0.6$$

$$P(A \text{ is selected, } B \text{ is not selected; } B \text{ is selected, } A \text{ is not selected}) = 0.6$$

$$P(A \cap B') + P(A' \cap B) = 0.6$$

$$\Rightarrow P(A)P(B') + P(A')P(B) = 0.6$$

$$\Rightarrow (0.7)(1 - p) + (0.3)p = 0.6$$

$$\Rightarrow p = 0.25$$

Thus, the probability that B gets selected is 0.25.

**2. The probability of simultaneous occurrence of at least one of two events A and B is  $p$ . If the probability that exactly one of A, B occurs is  $q$ , then prove that**

$$P(A') + P(B') = 2 - 2p + q.$$

**Ans:** Given,  $P(\text{exactly one of A, B occurs}) = q$ , we get

$$P(A \cup B) - P(A \cap B) = q$$

$$\Rightarrow p - P(A \cap B) = q$$

$$\Rightarrow P(A \cap B) = p - q$$

$$\Rightarrow 1 - P(A' \cup B') = p - q$$

$$\Rightarrow P(A' \cup B') = 1 - p + q$$

$$\Rightarrow P(A') + P(B') - P(A' \cap B') = 1 - p + q$$

$$\Rightarrow P(A') + P(B') = (1 - p + q) + P(A' \cap B')$$

$$\Rightarrow P(A') + P(B') = (1 - p + q) + (1 - P(A \cup B))$$

$$\Rightarrow P(A') + P(B') = (1 - p + q) + (1 - p)$$

$$\Rightarrow P(A') + P(B') = 2 - 2p + q$$

**3. 10% of the bulbs produced in a factory are of red colour and 2% are red and defective. If one bulb is picked up at random, determine the probability of its being defective if it is red.**

**Ans:** Let A and B be the events that the bulb is red and defective respectively.

$$P(A) = \frac{10}{100} = \frac{1}{10}$$

$$P(A \cap B) = \frac{2}{100} = \frac{1}{50}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{50} \times \frac{10}{1} = \frac{1}{5}$$

So, the probability of its being defective, if it is red, is  $\frac{1}{5}$ .

**4. Two dice are thrown together. Let A be the event 'getting 6 on the first die' and B be the event 'getting 2 on the second die'. Are the events A and B independent?**

**Ans:**  $A = \{(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

$B = \{(1,2), (2,2), (3,2), (4,2), (5,2), (6,2)\}$

$A \cap B = \{(6,2)\}$

$$P(A) = \frac{6}{36} = \frac{1}{6}, P(B) = \frac{1}{6}$$

$$P(A \cap B) = \frac{1}{36}$$

$A$  and  $B$  will be independent if,  $P(A \cap B) = P(A)P(B)$

$$LHS = P(A \cap B) = \frac{1}{36}$$

$$RHS = P(A)P(B) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

$$LHS = RHS$$

So,  $A$  and  $B$  are independent events.

**5. A committee of 4 students is selected at random from a group consisting 8 boys and 4 girls. Given that there is at least one girl in the committee, calculate the probability that there are exactly 2 girls in the committee.**

**Ans:** Let  $A$  denote the event that at least one girl will be chosen, and  $B$  the event that exactly 2 girls will be chosen. We require  $P(B|A)$ .

Because  $A$  denotes the event that at least one girl will be chosen,  $A'$  denotes that no girl is chosen, i.e., 4 boys are chosen. Then

$$\begin{aligned} P(A) &= \frac{{}^8C_4}{{}^{12}C_4} \\ &= \frac{70}{495} = \frac{14}{99} \left[ {}^nC_r = \frac{n!}{r!(n-r)!} \right] \end{aligned}$$

$$\text{Then, } P(A) = 1 - \frac{14}{99} = \frac{85}{99}$$

$$\text{Now } P(A \cap B) = P(2 \text{ boys and } 2 \text{ girls}) = \frac{{}^8C_2 \cdot {}^4C_2}{{}^{12}C_4}$$

$$= \frac{6 \times 28}{495} = \frac{56}{165}$$

$$\text{So, } P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{56}{165} \times \frac{99}{85} = \frac{168}{425}$$

**6. Three machines  $E_1, E_2, E_3$  in a certain factory produce 50%, 25% and 25%, respectively, of the total daily output of electric tubes. It is known that 4% of the tubes produced on each of machines  $E_1$  and  $E_2$  are defective, and that 5% of those produced on  $E_3$  are defective. If one tube is picked up at random from a day's production, calculate the probability that it is defective.**

**Ans:** Let D be the event that the picked up tube is defective.

Let  $A_1, A_2$  and  $A_3$  be the events that the tube is produced on machines  $E_1, E_2$  and  $E_3$ , respectively.

$$P(D) = P(A_1)P(D|A_1) + P(A_2)P(D|A_2) + P(A_3)P(D|A_3) \dots (1)$$

$$\text{Given, } P(A_1) = \frac{50}{100} = \frac{1}{2}, P(A_2) = \frac{1}{4}, P(A_3) = \frac{1}{4}$$

$$\text{Also } P(D|A_1) = P(D|A_2) = \frac{4}{100} = \frac{1}{25}$$

$$\text{And, } P(D|A_3) = \frac{5}{100} = \frac{1}{20}$$

Putting these values in (1), we get

$$P(D) = \frac{1}{2} \times \frac{1}{25} + \frac{1}{4} \times \frac{1}{25} + \frac{1}{4} \times \frac{1}{20}$$

$$= \frac{1}{50} + \frac{1}{100} + \frac{1}{80} = \frac{17}{400} = .0425$$

**7. Find the probability that in 10 throws of a fair die a score which is a multiple of 3 will be obtained in at least 8 of the throws.**

**Ans:** Given that, obtaining multiple of 3 in a throw of a die is a success.

There are total 10 throws of a fair die.

Let  $p$  be the probability of getting multiple of 3 in a throw of a die.

Since there can be a total 6 outcomes in a throw of a die. That is,  $\{1, 2, 3, 4, 5, 6\}$ .

And a multiple of 3 in a die is  $\{3\}, \{6\}$ .

$$\Rightarrow p = \frac{2}{6}$$

$$\Rightarrow p = \frac{1}{3}$$

Then, let  $q$  be the probability of not getting a multiple of 3 in a throw of a die.

And we know,  $p + q = 1$

$$\Rightarrow q = 1 - p$$

$$\Rightarrow q = 1 - \frac{1}{3}$$

$$\Rightarrow q = \frac{3-1}{3}$$

$$\Rightarrow q = \frac{2}{3}$$

Let  $X$  be a random variable representing a number of successes (getting multiple of 3 in die) out of 10 throws of a die.

Then, the probability of getting  $r$  successes out of  $n$  throws of die is given by,

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

Here  $n=10$ .

Now, put values of  $n$ ,  $p$ , and  $q$  in the above equation.

$$P(X = r) = {}^{10}C_r \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{10-r} \dots (i)$$

We need to find the probability of getting a multiple of 3 in at least 8 of the throws out of 10 throws of a fair die.

It is given,

$$\text{Probability} = P(X \geq 8)$$

This can be written as,  $P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10)$

Just put  $r = 8, 9, 10$  in equation (i) to find the value of  $P(X = 8)$ ,  $P(X = 9)$ ,  $P(X = 10)$  respectively, then substitute in the above equation.

$$\Rightarrow P(X \geq 8) = {}^{10}C_8 \left(\frac{1}{3}\right)^8 \left(\frac{2}{3}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{3}\right)^9 \left(\frac{2}{3}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{3}\right)^{10} \left(\frac{2}{3}\right)^{10-10}$$

$$\Rightarrow P(X \geq 8) = {}^{10}C_8 \left(\frac{1}{3}\right)^8 \left(\frac{2}{3}\right)^2 + {}^{10}C_9 \left(\frac{1}{3}\right)^9 \left(\frac{2}{3}\right)^1 + {}^{10}C_{10} \left(\frac{1}{3}\right)^{10} \left(\frac{2}{3}\right)^0$$

$$\Rightarrow P(X \geq 8) = \left[ \left( \frac{10!}{(10-8)!8!} \right) \times \left( \frac{2^2}{3^{10}} \right) \right] + \left[ \left( \frac{10!}{(10-9)!9!} \right) \times \left( \frac{2}{3^{10}} \right) \right] + \left[ \left( \frac{10!}{(10-10)!10!} \right) \times \left( \frac{1}{3} \right)^{10} \right]$$

$$\Rightarrow P(X \geq 8) = \left[ \left( \frac{10!}{2!8!} \right) \times 4 \times \left( \frac{1}{3} \right)^{10} \right] + \left[ \left( \frac{10!}{9!} \right) \times 2 \times \left( \frac{1}{3} \right)^{10} \right] + \left[ \left( \frac{10!}{10!} \right) \times \left( \frac{1}{3} \right)^{10} \right]$$

$$\Rightarrow P(X \geq 8) = \left( \frac{1}{3} \right)^{10} \left[ \left( \frac{10 \times 9 \times 8!}{2 \times 8!} \times 4 \right) + \left( \frac{10 \times 9!}{9!} \times 2 \right) + 1 \right]$$

$$\Rightarrow P(X \geq 8) = \left( \frac{1}{3} \right)^{10} [180 + 20 + 1]$$

$$\Rightarrow P(X \geq 8) = 201 \times \left(\frac{1}{3}\right)^{10}$$

$$\Rightarrow P(X \geq 8) = \frac{201}{3^{10}}$$

**8. A discrete random variable X has the following probability distribution:**

|             |          |           |           |           |                      |                       |                           |
|-------------|----------|-----------|-----------|-----------|----------------------|-----------------------|---------------------------|
| <b>X</b>    | <b>1</b> | <b>2</b>  | <b>3</b>  | <b>4</b>  | <b>5</b>             | <b>6</b>              | <b>7</b>                  |
| <b>P(X)</b> | <b>C</b> | <b>2C</b> | <b>2C</b> | <b>3C</b> | <b>C<sup>2</sup></b> | <b>2C<sup>2</sup></b> | <b>7C<sup>2</sup> + C</b> |

**Find the value of C. Also find the mean of the distribution.**

**Ans:** Since  $\sum p_i = 1$ , we have

$$C + 2C + 2C + 3C + C^2 + 2C^2 + 7C^2 + C = 1$$

$$\text{or, } 10C^2 + 9C - 1 = 0$$

$$\text{or, } (10C - 1)(C + 1) = 0$$

$$\text{Either } (10C - 1) = 0 \text{ or } (C + 1) = 0$$

$$\Rightarrow C = \frac{1}{10}, C = -1$$

$C = -1$  cannot be possible. probability cannot be negative.

So, the permissible value of  $C = \frac{1}{10}$ .

$$\text{Mean} = \sum_{i=1}^n x_i p_i = \sum_{i=1}^7 x_i p_i$$

$$\sum_{i=1}^7 x_i p_i = 1 \times \frac{1}{10} + 2 \times \frac{2}{10} + 3 \times \frac{2}{10} + 4 \times \frac{3}{10} + 5 \left(\frac{1}{10}\right)^2 + 6 \times 2 \left(\frac{1}{10}\right)^2 + 7 \left(7 \left(\frac{1}{10}\right)^2 + \frac{1}{10}\right)$$

$$\sum_{i=1}^7 x_i p_i = \frac{1}{10} + \frac{4}{10} + \frac{6}{10} + \frac{12}{10} + \frac{5}{100} + \frac{12}{100} + \frac{49}{100} + \frac{7}{10}$$

$$\sum_{i=1}^7 x_i p_i = 3.66.$$

### Long Answer Type Question

**9. Four balls are to be drawn without replacement from a box containing 8 red and 4 white balls. If  $X$  denotes the number of red balls drawn, find the probability distribution of  $X$ .**

**Ans:** Since 4 balls have to be drawn, therefore,  $X$  can take the values 0,1,2,3,4.

$$P(X = 0) = P(\text{no red ball}) = P(4 \text{ white balls})$$

$$= \frac{{}^4C_4}{{}^{12}C_4} = \frac{1}{495}$$

$$P(X = 1) = P(1 \text{ red ball and 3 white balls})$$

$$= \frac{{}^8C_1 \times {}^4C_3}{{}^{12}C_4} = \frac{32}{495}$$

$$P(X = 2) = P(2 \text{ red balls and 2 white balls})$$

$$= \frac{{}^8C_2 \times {}^4C_2}{{}^{12}C_4} = \frac{168}{495}$$

$$P(X = 3) = P(3 \text{ red balls and 1 white ball})$$

$$= \frac{{}^8C_3 \times {}^4C_1}{{}^{12}C_4} = \frac{224}{495}$$

$$P(X = 4) = P(4 \text{ red balls}) = \frac{{}^8C_4}{{}^{12}C_4} = \frac{70}{495}$$

So, the following is the required probability distribution of  $X$

|     |   |   |   |   |   |
|-----|---|---|---|---|---|
| $X$ | 0 | 1 | 2 | 3 | 4 |
|-----|---|---|---|---|---|

|        |                 |                  |                   |                   |                  |
|--------|-----------------|------------------|-------------------|-------------------|------------------|
| $P(X)$ | $\frac{1}{495}$ | $\frac{32}{495}$ | $\frac{168}{495}$ | $\frac{224}{495}$ | $\frac{70}{495}$ |
|--------|-----------------|------------------|-------------------|-------------------|------------------|

**10. Determine variance and standard deviation of the number of heads in three tosses of a coin.**

**Ans:** Let  $X$  denote the number of heads tossed. So,  $X$  can take the values 0, 1, 2, 3. When a coin is tossed three times, we get

Sample space  $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$$P(X = 0) = P(\text{no head}) = P(TTT) = \frac{1}{8}$$

$$P(X = 1) = P(\text{one head}) = P(HTT, THT, TTH) = \frac{3}{8}$$

$$P(X = 2) = P(\text{two heads}) = P(HHT, HTH, THH) = \frac{3}{8}$$

$$P(X = 3) = P(\text{three heads}) = P(HHH) = \frac{1}{8}$$

So, the probability distribution of  $X$  is:

|        |               |               |               |               |
|--------|---------------|---------------|---------------|---------------|
| $X$    | 0             | 1             | 2             | 3             |
| $P(X)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ |

$$\text{Variance of } X = \sigma^2 = \sum x_i^2 p_i - \mu^2 \dots (1)$$

where  $\mu$  is the mean of  $X$  given by

$$\mu = \sum x_i p_i = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8}$$

$$= \frac{3}{2} \dots \dots (2)$$

$$\text{Now } \sum x_i^2 p_i = 0^2 \times \frac{1}{8} + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 3^2 \times \frac{1}{8} = 3 \dots (3)$$

From (1), (2) and (3), we get

$$\sigma^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$$

$$\text{And standard deviation} = \sqrt{\sigma^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}.$$

**11. Refer to Example 6. Calculate the probability that the defective tube was produced on machine  $E_1$ .**

**Ans:** Now, we have to find  $P(A_1 / D)$ .

$$P(A_1 / D) = \frac{P(A_1 \cap D)}{P(D)} = \frac{P(A_1)P(D / A_1)}{P(D)}$$

$$P(A_1 / D) = \frac{\frac{1}{2} \times \frac{1}{25}}{\frac{17}{400}} = \frac{8}{17}$$

**12. A car manufacturing factory has two plants, X and Y. Plant X manufactures 70% of cars and plant Y manufactures 30%. 80% of the cars at plant X and 90% of the cars at plant Y are rated of standard quality. A car is chosen at random and is found to be of standard quality. What is the probability that it has come from plant X?**

**Ans:** Let E be the event that the car is of standard quality. Let  $B_1$  and  $B_2$  be the events that the car is manufactured in plants X and Y, respectively.

$$P(B_1) = \frac{70}{100} = \frac{7}{10}, P(B_2) = \frac{30}{100} = \frac{3}{10}$$

$P(E|B_1)$  = Probability that a standard quality car is manufactured in plant  $X$

$$P(E|B_1) = \frac{80}{100} = \frac{8}{10}$$

$$\text{And, } P(E|B_2) = \frac{90}{100} = \frac{9}{10}$$

$P(B_1|E)$  = Probability that a standard quality car has come from plant  $X$

$$P(B_1|E) = \frac{P(B_1) \times P\left(\frac{E}{B_1}\right)}{P(B_1) \cdot P\left(\frac{E}{B_1}\right) + P(B_2) \cdot P\left(\frac{E}{B_2}\right)}$$

$$P(B_1|E) = \frac{\frac{7}{10} \times \frac{8}{10}}{\frac{7}{10} \times \frac{8}{10} + \frac{3}{10} \times \frac{9}{10}} = \frac{56}{83}$$

Hence, the required probability is  $\frac{56}{83}$ .

### Objective Type Questions

Choose the correct answer from the given four options in each of the Examples 13 to 17.

13. Let  $A$  and  $B$  be two events. If  $P(A) = 0.2, P(B) = 0.4, P(A \cup B) = 0.6$ , then  $P(A|B)$  is equal to

(A) 0.8

(B) 0.5

(C) 0.3

(D) 0

**Ans:** Correct answer - D

From the given data  $P(A) + P(B) = P(A \cup B)$ .

This means,  $P(A \cap B) = 0$ . Thus,  $P(A|B) = \frac{P(A \cap B)}{P(B)} = 0$

**14. Let A and B be two events such that  $P(A) = 0.6$ ,  $P(B) = 0.2$ , and  $P(A|B) = 0.5$ .**

**Then  $P(A'|B')$  equals**

(A)  $\frac{1}{10}$

(B)  $\frac{3}{10}$

(C)  $\frac{3}{8}$

(D)  $\frac{6}{7}$

**Ans:** Correct answer - C

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B) = 0.5 \times 0.2 = 0.1$$

$$P(A'|B') = \frac{P(A' \cap B')}{P(B')} = \frac{1 - P(A \cup B)}{1 - P(B)}$$

$$P(A'|B') = \frac{1 - P(A) - P(B) + P(A \cap B)}{1 - 0.2} = \frac{3}{8}$$

**15. If A and B are independent events such that  $0 < P(A) < 1$  and  $0 < P(B) < 1$ , then which of the following is not correct?**

**(A) A and B are mutually exclusive**

**(B) A and B' are independent**

**(C) A and B are independent**

**(D) A and B' are independent**

**Ans:** Correct answer - A

We have A and B are independent events,

$$\text{Hence } P\left(\frac{A}{B}\right) = P(A)$$

$$\text{Hence } \frac{P(A \cap B)}{P(B)} = P(A)$$

Multiplying both sides by  $P(B)$ , we get

$$P(A \cap B) = P(A)P(B)$$

Two events A and B are said to be mutually exclusive if the sets A and B are disjoint.

Consider two events A and B such that A has non-zero probability and  $B = S$ .

Since  $A \subset S$ , we have  $A \cap B = A \neq \phi$ . Hence the events are not mutually exclusive.

However,  $P(A \cap B) = P(A)$  and  $P(A)P(B) = P(A)$

Hence the events are independent but not mutually exclusive. Hence option a is incorrect.

Now if A and B are independent then,  $P(A \cap B) = P(A)P(B)$

Now we know that  $A = A \cap (B \cup B') = (A \cap B) \cup (A \cap B')$

Hence  $P(A) = P((A \cap B) \cup (A \cap B'))$

Since the events  $(A \cap B)$  and  $(A \cap B')$  are disjoint, these events are mutually exclusive

Hence  $P(A) = P((A \cap B) \cup (A \cap B')) = P(A \cap B) + P(A \cap B')$

Hence, we have

$$P(A) = P(A)P(B) + P(A \cap B')$$

$$\Rightarrow P(A \cap B') = P(A)(1 - P(B))$$

$$\text{We know that } P(B') = 1 - P(B)$$

$$\text{Hence we have } P(A \cap B') = P(A)P(B')$$

Hence the events  $A$  and  $B'$  are independent,

Since  $A$  and  $B'$  are independent, we have  $A'$  and  $B'$  are also independent.

Hence options  $b$  and  $c$  are correct.

$$\text{Now we know that } P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Hence, we have

$$P(A/B) + P(A'/B) = \frac{P(A \cap B)}{P(B)} + \frac{P(A' \cap B)}{P(B)} = \frac{P(A \cap B) + P(A' \cap B)}{P(B)}$$

Since  $A \cap B$  and  $A' \cap B$  are mutually exclusive events, we have

$$P(A \cap B) + P(A' \cap B) = P((A \cap B) \cup (A' \cap B)) = P((A \cup A') \cap B) = P(B)$$

Hence, we have

$$P(A/B) + P(A'/B) = \frac{P(B)}{P(B)} = 1$$

Hence option  $d$  is correct.

Therefore, the option that is incorrect is option  $A$ .

**16. Let  $X$  be a discrete random variable. The probability distribution of  $X$  is given below:**

|             |               |                |               |
|-------------|---------------|----------------|---------------|
| <b>X</b>    | <b>30</b>     | <b>10</b>      | <b>-10</b>    |
| <b>P(X)</b> | $\frac{1}{5}$ | $\frac{3}{10}$ | $\frac{1}{2}$ |

Then  $E(X)$  is equal to

- (A) 6
- (B) 4
- (C) 3
- (D) -5

**Ans:** Correct answer - B

Because we know that  $E(X) = \sum_{i=1}^n x_i p_i$

$$\text{So, } E(X) = 30 \times \frac{1}{5} + 10 \times \frac{3}{10} - 10 \times \frac{1}{2} = 4$$

**17. Let  $X$  be a discrete random variable assuming values  $x_1, x_2, \dots, x_n$  with probabilities  $P_1, P_2, \dots, P_n$ , respectively. Then variance of  $X$  is given by**

- (A)  $E(X^2)$
- (B)  $E(X^2) + E(X)$
- (C)  $E(X^2) - [E(X)]^2$
- (D)  $\sqrt{E(X^2) - [E(X)]^2}$

**Ans:** Correct answer - C

As we know that probability of  $X$  is equal to mean ( $\mu$ ) of  $X$ .

$\therefore$  The probability of  $X$  and  $X^2$  is

$$\mu = E(X) = \sum_{i=1}^n x_i p_i$$

$$E(X^2) = \sum_{i=1}^n x_i^2 p_i$$

The variance of  $X$  ( $\sigma$ ) when it is discrete is given by the addition of square of the mean of  $X$  ( $\mu$ ) and the probability of  $X^2$ .

$$\sigma^2 = \sum_{i=1}^n (x_i - \mu)^2 p_i = \sum_{i=1}^n x_i^2 p_i - \mu^2 \text{ or } \sigma = E(X - \mu)^2$$

The standard deviation of the random variable  $X$  is defined as  $\sigma^2 = \sqrt{\text{variance}(X)}$

$$\text{variance}(X) = \sum_{i=1}^n x_i^2 p_i + \left[ \sum_{i=1}^n x_i p_i \right]^2$$

$$\text{variance}(X) = E(X^2) + [E(X)]^2$$

**Fill in the blanks in Examples 18 and 19**

**18. If A and B are independent events such that  $P(A) = p, P(B) = 2p$  and  $P(\text{Exactly one of } A, B) = \frac{5}{9}$ , then  $p = \dots$**

$$\text{Ans: } p = \frac{1}{3} = \frac{5}{12} \left[ (1-p)(2p) + p(1-2p) = 3p - 4p^2 = \frac{5}{9} \right]$$

**19. If A and  $B'$  are independent events then  $P(A' \cup B) = 1 - \dots$**

$$\text{Ans: } P(A' \cup B) = 1 - P(A \cap B') = 1 - P(A)P(B')$$

(since A and  $B'$  are independent).

**State whether each of the statement in Examples 20 to 22 is True or False**

**20. Let A and B be two independent events. Then  $P(A \cap B) = P(A) + P(B)$ .**

**Ans:** False, because  $P(A \cap B) = P(A)P(B)$  when events A and B are independent.

**21. Three events A, B and C are said to be independent if**

**$P(A \cap B \cap C) = P(A)P(B)P(C)$ .**

**Ans:** False. Reason is that A,B,C will be independent if they are pairwise independent and  $P(A \cap B \cap C) = P(A)P(B)P(C)$ .

**22. One of the conditions of Bernoulli trials is that the trials are independent of each other.**

**Ans:** True.

## Exercise

### Short Answer Type Questions

**1. For a loaded die, the probabilities of outcomes are given as under:**

**$P(1) = P(2) = 0.2, P(3) = P(5) = P(6) = 0.1$  and  $P(4) = 0.3$ .**

**The die is thrown two times. Let A and B be the events, 'same number each time' and 'a total score is 10 or more', respectively. Determine whether or not A and B are independent.**

**Ans:** Given

$$P(1) = P(2) = 0.2,$$

$$P(3) = P(5) = P(6) = 0.1$$

$$\text{and } P(4) = 0.3$$

Die is thrown two times

A = Same number each time and B = Total score is 10 or more

$$\begin{aligned}\text{So, } P(A) &= [P(1,1) + P(2,2) + P(3,3) + P(4,4) + P(5,5) + P(6,6)] \\ &= [P(1) \cdot P(1) + P(2) \cdot P(2) + P(3) \cdot P(3) + P(4) \cdot P(4) + P(5) \cdot P(5) + P(6) \cdot P(6)] \\ &= [0.2 \times 0.2 + 0.2 \times 0.2 + 0.1 \times 0.1 + 0.3 \times 0.3 + 0.1 \times 0.1 + 0.1 \times 0.1] \\ &= 0.04 + 0.04 + 0.01 + 0.09 + 0.01 + 0.01 = 0.20\end{aligned}$$

$$\text{Now } B = \{(4,6), (6,4), (5,5), (5,6), (6,5), (6,6)\}$$

$$\begin{aligned}P(B) &= P(4,6) + P(6,4) + P(5,5) + P(5,6) + P(6,5) + P(6,6) \\ &= P(4)P(6) + P(6)P(4) + P(5)P(5) + P(5)P(6) + P(6)P(5) + P(6)P(6) \\ &= 0.3 \times 0.1 + 0.1 \times 0.3 + 0.1 \times 0.1 + 0.1 \times 0.1 + 0.1 \times 0.1 + 0.1 \times 0.1 \\ &= 0.03 + 0.03 + 0.01 + 0.01 + 0.01 + 0.01 = 0.10\end{aligned}$$

$$\text{And, } A \cap B = \{(5,5), (6,6)\}$$

$$\begin{aligned}\therefore P(A \cap B) &= P(5,5) + P(6,6) = P(5)P(5) + P(6)P(6) \\ &= 0.1 \times 0.1 + 0.1 \times 0.1 = 0.01 + 0.01 = 0.02\end{aligned}$$

$A$  and  $B$  both will be independent events, if

$$P(A \cap B) = P(A)P(B).$$

$$\text{Here, } P(A \cap B) = 0.02 \text{ and } P(A)P(B) = 0.20 \times 0.10 = 0.02$$

$$\text{So, } P(A \cap B) = P(A) \cdot P(B) = 0.02$$

Hence,  $A$  and  $B$  are independent events.

**2. Refer to Exercise 1 above. If the die were fair, determine whether or not the events  $A$  and  $B$  are independent.**

**Ans:** We have

$$A = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$$

$$n(A) = 6 \text{ and } n(S) = 6^2 = 36$$

$$\text{So, } P(A) = \frac{n(A)}{n(B)} = \frac{6}{36} = \frac{1}{6}$$

$$\text{and } B = \{(4,6), (6,4), (5,5), (6,5), (5,6), (6,6)\}$$

$$n(B) = 6 \text{ and } n(S) = 6^2 = 36$$

$$\text{So, } P(B) = \frac{n(B)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

$$A \cap B = \{(5,5), (6,6)\}$$

$$n(A \cap B) = 2 \text{ and } n(S) = 36$$

$$\text{So, } (A \cap B) = \frac{2}{36} = \frac{1}{18}$$

$$\text{and, } P(A) \cdot P(B) = \frac{1}{36}$$

$$\text{So, } P(A \cap B) \neq P(A) \cdot P(B)$$

Hence,  $A$  and  $B$  are not independent events.

**3. The probability that at least one of the two events  $A$  and  $B$  occurs is 0.6. If  $A$  and  $B$  occur simultaneously with probability 0.3, evaluate  $P(\bar{A}) + P(\bar{B})$ .**

**Ans:** We know that,  $A \cup B$  denotes the occurrence of at least one of  $A$  and  $B$  and  $A \cap B$  denotes the occurrence of both  $A$  and  $B$ , simultaneously.

$$\text{Thus, } P(A \cup B) = 0.6 \text{ and } P(A \cap B) = 0.3$$

$$\text{Also, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 0.6 = P(A) + P(B) - 0.3$$

$$\Rightarrow P(A) + P(B) = 0.9$$

$$\Rightarrow [1 - P(\bar{A})] + [1 - P(\bar{B})] = 0.9 [\because P(A) = 1 - P(\bar{A}) \text{ and } P(B) = 1 - P(\bar{B})]$$

$$\Rightarrow P(\bar{A}) + P(\bar{B}) = 2 - 0.9 = 1.1$$

**4. A bag contains 5 red marbles and 3 black marbles. Three marbles are drawn one by one without replacement. What is the probability that at least one of the three marbles drawn be black, if the first marble is red?**

**Ans:** Let  $R$  stand for red marbles and  $B$  is for black.

If at least one of the three marbles drawn be black, if the first marble is red.

(i)  $E_1$  = Second ball is black and third ball is red .

(ii)  $E_2$  = Second ball is black and third ball is also black .

(iii)  $E_3$  = Second ball is red and third ball is black .

$$\therefore P(E_1) = P(R_1) \cdot P\left(\frac{B_1}{R_1}\right) P\left(\frac{R_2}{R_1 B_1}\right) = \frac{5}{8} \cdot \frac{3}{7} \cdot \frac{4}{6} = \frac{60}{336} = \frac{5}{28}$$

$$P(E_2) = P(R_1) P\left(\frac{B_1}{R_1}\right) P\left(\frac{B_2}{R_1 B_1}\right) = \frac{5}{8} \cdot \frac{3}{7} \cdot \frac{2}{6} = \frac{30}{336} = \frac{5}{56}$$

$$\text{And } P(E_3) = P(R_1) \cdot P\left(\frac{R_2}{R_1}\right) P\left(\frac{B_1}{R_1 R_2}\right) = \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} = \frac{60}{336} = \frac{5}{28}$$

$$\therefore P(E) = P(E_1) + P(E_2) + P(E_3) = \frac{5}{28} + \frac{5}{56} + \frac{5}{28}$$

$$P(E) = \frac{10 + 5 + 10}{56} = \frac{25}{56}$$

$$\text{So, the probability} = \frac{25}{56}$$

**5. Two dice are thrown together and the total score is noted. The events E, F and G are 'a total of 4', 'a total of 9 or more', and 'a total divisible by 5',**

respectively. Calculate  $P(E)$ ,  $P(F)$  and  $P(G)$  and decide which pairs of events, if any, are independent.

**Ans:** Two dice are thrown together

$$SO, n(S) = 36$$

$$E = \text{A total of } 4 = \{(2, 2), (3, 1), (1, 3)\}$$

$$\text{so, } n(E) = 3$$

and  $F = \text{A total of } 9 \text{ or more}$

$$= \{(3, 6), (6, 3), (4, 5), (5, 4), (6, 4), (4, 6), (5, 5), (6, 5), (5, 6), (6, 6)\}$$

$$\text{so, } n(F) = 10$$

and  $G = \text{a total divisible by } 5 = \{(1, 4), (4, 1), (2, 3), (3, 2), (4, 6), (6, 4), (5, 5)\}$

$$\text{so, } n(G) = 7$$

Here, we see that  $(E \cap F) = \phi$  and  $(E \cap G) = \phi$

$$\text{and } (F \cap G) = \{(4, 6), (6, 4), (5, 5)\}$$

$$\Rightarrow n(F \cap G) = 3 \text{ and } (E \cap F \cap G) = \phi$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

$$P(F) = \frac{n(F)}{n(S)} = \frac{10}{36} = \frac{5}{18}$$

$$P(G) = \frac{n(G)}{n(S)} = \frac{7}{36}$$

$$P(F \cap G) = \frac{3}{36} = \frac{1}{12}$$

$$\text{and } P(F) \cdot P(G) = \frac{5}{18} \cdot \frac{7}{36} = \frac{35}{648}$$

Since,  $P(F \cap G) \neq P(F) \cdot P(G)$

Hence, there is no pair which is independent.

**6. Explain why the experiment of tossing a coin three times is said to have binomial distribution.**

**Ans:** We know that, a random variable  $X$  takes values  $0, 1, 2, \dots, n$  is said to be binomial distribution having parameters  $n$  and  $p$ , if its probability is given by

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

Where  $q = 1 - p$ , and  $r = 0, 1, 2, \dots, n$

Similarly, in case tossing a coin 3 times,

$n = 3$  and  $X$  has values  $r = 0, 1, 2$  and 3 with  $p = \frac{1}{2}$  and  $q = \frac{1}{2}$ .

Hence, it is said to have a binomial distribution.

**7. If A and B are two events such that  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{1}{3}$  and  $P(A \cap B) = \frac{1}{4}$ .**

**Find:**

**(i)  $P(A|B)$**

**Ans:** We have,  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{1}{3}$  and  $P(A \cap B) = \frac{1}{4}$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{3}} = \frac{3}{4}$$

**(ii)  $P(B|A)$**

**Ans:** We have,  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{1}{3}$  and  $P(A \cap B) = \frac{1}{4}$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

(iii)  $P(A' | B)$

$$\text{Ans: } P\left(\frac{A'}{B}\right) = 1 - P\left(\frac{A}{B}\right) = 1 - \frac{3}{4} = \frac{1}{4}$$

(iv)  $P(A' | B')$

$$\begin{aligned} \text{Ans: } P\left(\frac{A'}{B'}\right) &= \frac{P(A' \cap B')}{P(B')} = \frac{1 - P(A \cup B)}{1 - P(B)} \\ &= \frac{1 - [P(A) + P(B) - P(A \cap B)]}{1 - P(B)} \\ &= \frac{1 - \left[\frac{1}{2} + \frac{1}{3} - \frac{1}{4}\right]}{1 - \frac{1}{3}} = \frac{1 - \frac{14}{24}}{\frac{2}{3}} = \frac{\frac{10}{24}}{\frac{2}{3}} = \frac{5}{8} \end{aligned}$$

8. Three events A, B and C have probabilities  $\frac{2}{5}$ ,  $\frac{1}{3}$  and  $\frac{1}{2}$  respectively. Given that  $P(A \cap B) = \frac{1}{5}$  and  $P(B \cap C) = \frac{1}{4}$  find the value of  $P(C | B)$  and  $P(A' \cap C')$ .

$$\text{Ans: We have, } P(A) = \frac{2}{5}, P(B) = \frac{1}{3}, P(C) = \frac{1}{2}$$

$$P(A \cap C) = \frac{1}{5} \text{ and } P(B \cap C) = \frac{1}{4}$$

$$\therefore P(C / B) = \frac{P(B \cap C)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{3}} = \frac{3}{4}$$

$$P(A' \cap C') = 1 - P(A \cup C) = 1 - [P(A) + P(C) - P(A \cap C)]$$

$$P(A' \cap C') = 1 - \left[ \frac{2}{5} + \frac{1}{2} - \frac{1}{5} \right] = 1 - \left[ \frac{4+5-2}{10} \right] = 1 - \frac{7}{10} = \frac{3}{10}$$

**9. Let  $E_1$  and  $E_2$  be two independent events such that  $P(E_1) = p_1$  and  $P(E_2) = p_2$ .**

**Describe in words of the events whose probabilities are:**

**(i)  $p_1 p_2$**

**Ans:** Here,  $P(E_1) = p_1$  and  $P(E_2) = p_2$

$$\text{So, } p_1 p_2 \Rightarrow P(E_1) \cdot P(E_2) = P(E_1' \cap E_2)$$

So,  $E_1$  and  $E_2$  occur.

**(ii)  $(1 - p_1)p_2$**

$$\text{Ans: } (1 - p_1)p_2 = P(E_1') P(E_2) = P(E_1' \cap E_2)$$

So,  $E_1$  does not occur but  $E_2$  occurs.

**(iii)  $1 - (1 - p_1)(1 - p_2)$**

$$\text{Ans: } 1 - (1 - p_1)(1 - p_2) = 1 - P(E_1') P(E_2') = 1 - P(E_1' \cap E_2')$$

$$= 1 - [1 - P(E_1 \cup E_2)] = P(E_1 \cup E_2)$$

So, either  $E_1$  or  $E_2$  or both  $E_1$  and  $E_2$  occurs.

(iv)  $P_1 + P_2 - 2P_1P_2$

**Ans:**  $P_1 + P_2 - 2P_1P_2 = P(E_1) + P(E_2) - 2P(E_1) \cdot P(E_2)$

$$= P(E_1) + P(E_2) - 2P(E_1 \cap E_2)$$

$$= P(E_1 \cup E_2) - P(E_1 \cap E_2)$$

So, either  $E_1$  or  $E_2$  occurs but not both.

**10. A discrete random variable X has the probability distribution given as below:**

|             |            |                         |                          |          |
|-------------|------------|-------------------------|--------------------------|----------|
| <b>X</b>    | <b>0.5</b> | <b>1</b>                | <b>1.5</b>               | <b>2</b> |
| <b>P(X)</b> | <b>k</b>   | <b><math>k^2</math></b> | <b><math>2k^2</math></b> | <b>k</b> |

**(i) Find the value of k**

**Ans:**  $\sum_{i=1}^n P_i = 1$ , where  $P_i \geq 0$

$$\Rightarrow P_1 + P_2 + P_3 + P_4 = 1$$

$$\Rightarrow k + k^2 + 2k^2 + k = 1$$

$$\Rightarrow 3k^2 + 2k - 1 = 0$$

$$\Rightarrow 3k^2 + 3k - k - 1 = 0$$

$$\Rightarrow 3k(k+1) - 1(k+1) = 0$$

$$\Rightarrow (3k-1)(k+1) = 0$$

$$\Rightarrow k = \frac{1}{3} \text{ and } k = -1$$

but  $k$  is  $\geq 0 \Rightarrow k = \frac{1}{3}$

**(ii) Determine the mean of the distribution.**

**Ans:** Mean of the distribution

$$\begin{aligned} E(X) &= \sum_{i=1}^n x_i p_i \\ &= 0.5(k) + 1(k^2) + 1.5(2k^2) + 2k = 4k^2 + 2.5k \\ &= 4 \cdot \frac{1}{9} + 2.5 \cdot \frac{1}{3} \left[ \because k = \frac{1}{3} \right] \\ &= \frac{4 + 7.5}{9} = \frac{23}{18} \end{aligned}$$

**11. Prove that**

**(i)  $P(A) = P(A \cap B) + P(A \cap \bar{B})$**

**Ans:** To prove  $P(A) = P(A \cap B) + P(A \cap \bar{B})$

$$\begin{aligned} RHS &= P(A \cap B) + P(A \cap \bar{B}) \\ &= P(A) \cdot P(B) + P(A) \cdot P(\bar{B}) \\ &= P(A)[P(B) + P(\bar{B})] \\ &= P(A)[P(B) + 1 - P(B)] \quad [\because P(\bar{B}) = 1 - P(B)] \\ &= P(A) = LHS \text{ Hence proved.} \end{aligned}$$

**(ii)  $P(A \cup B) = P(A \cap B) + P(A \cap \bar{B}) + P(\bar{A} \cap B)$**

**Ans:** To prove,  $P(A \cup B) = P(A \cap B) + P(A \cap \bar{B}) + P(\bar{A} \cap B)$

$$\begin{aligned} RHS &= P(A) \cdot P(B) + P(A) \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B) \\ &= P(A) \cdot P(B) + P(A) \cdot [1 - P(B)] + [1 - P(A)]P(B) \\ &= P(A) \cdot P(B) + P(A) - P(A) \cdot P(B) + P(B) - P(A) \cdot P(B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \end{aligned}$$

$$= P(A) + P(B) - P(A \cap B)$$

$$= P(A \cup B) = LHS$$

**12. If  $X$  is the number of tails in three tosses of a coin, determine the standard deviation of  $X$ .**

**Ans:** Given that,  $X = 0, 1, 2, 3$

$$\Rightarrow P(X = r) = {}^n C_r (p)^r (q)^{n-r}$$

$$\text{where } n = 3, p = \frac{1}{2}, q = \frac{1}{2}$$

$$\text{and } n = 3, p = \frac{1}{2}, q = \frac{1}{2}, r = 0, 1, 2, 3$$

|           |               |               |               |               |
|-----------|---------------|---------------|---------------|---------------|
| $X$       | 0             | 1             | 2             | 3             |
| $P(X)$    | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ |
| $XP(X)$   | 0             | $\frac{3}{4}$ | $\frac{3}{8}$ | $\frac{3}{8}$ |
| $X^2P(X)$ | 0             | $\frac{3}{8}$ | $\frac{3}{2}$ | $\frac{9}{8}$ |

We know that,  $\text{Var}(X) = E(X^2) - [E(X)]^2$  Eq...(i)

$$\text{Where, } E(X^2) = \sum_{i=1}^n x_i^2 P(x_i) \text{ and } E(X) = \sum_{i=1}^n x_i P(x_i)$$

$$\therefore E(X^2) = \sum_{i=1}^n x_i^2 P(x_i) = 0 + \frac{3}{8} + \frac{3}{2} + \frac{9}{8} = \frac{24}{8} = 3$$

$$\text{And } [E(X)]^2 = \left[ \sum_{i=1}^n x_i P(x_i) \right]^2 = \left[ 0 + \frac{3}{8} + \frac{3}{4} + \frac{3}{8} \right]^2 = \left[ \frac{12}{8} \right]^2 = \frac{9}{4}$$

$$\text{Var}(X) = 3 - \frac{9}{4} = \frac{3}{4} \text{ [ using Eq. (i) ]}$$

$$\text{Standard deviation of } X = \sqrt{\text{Var}(X)} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

**13. In a dice game, a player pays a stake of Re1 for each throw of a die. She receives Rs 5 if the die shows a 3, Rs 2 if the die shows a 1 or 6, and nothing otherwise. What is the player's expected profit per throw over a long series of throws?**

**Ans:** Let  $X$  be the random variable of profit per throw.

|        |               |               |               |
|--------|---------------|---------------|---------------|
| $X$    | -1            | 1             | 4             |
| $P(X)$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{6}$ |

Since, she loses Rs 1 for getting any of 2, 4 or 5 .

$$\text{so, } P(X = -1) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

$$P(X = 1) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \text{ [die shows 1 or 6]}$$

$$P(X = 4) = \frac{1}{6} \text{ [ die shows a 3]}$$

$$\text{Player's expected profit} = \sum X P(X)$$

$$= -1 \times \frac{1}{2} + 1 \times \frac{1}{3} + 4 \times \frac{1}{6}$$

$$= \frac{-3 + 2 + 4}{6} = \frac{3}{6} = \frac{1}{2} = 0.50$$

**14. Three dice are thrown at the same time. Find the probability of getting three two's, if it is known that the sum of the numbers on the dice was six.**

**Ans:** The dice is thrown three times

sample space  $[n(S)] = 6^3 = 216$

Let  $E_1$  = event of sum of numbers on the dice was six

and  $E_2$  = event of three two's.

$$\Rightarrow n(E_1) = 10$$

$$E_1 = \{(1,1,4), (1,2,3), (1,3,2), (1,4,1), (2,1,3), (2,2,2), (2,3,1), (3,1,2), (3,2,1), (4,1,1)\}$$

$$\text{and } E_2 = \{2,2,2\}$$

$$\Rightarrow n(E_2) = 1$$

$$\text{Also, } (E_1 \cap E_2) = 1$$

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{\frac{1}{216}}{\frac{10}{216}} = \frac{1}{10}$$

**15. suppose 10,000 tickets are sold in a lottery each for Re 1 . First prize is of Rs 3000 and the second prize is of Rs. 2000 . There are three third prizes of Rs. 500 each. If you buy one ticket, what is your expectation?**

**Ans:** Let  $X$  be the random variable where  $X = 0, 500, 2000 \text{ and } 3000$

|        |                      |                   |                   |                   |
|--------|----------------------|-------------------|-------------------|-------------------|
| $X$    | 0                    | 500               | 2000              | 3000              |
| $P(X)$ | $\frac{9995}{10000}$ | $\frac{3}{10000}$ | $\frac{1}{10000}$ | $\frac{1}{10000}$ |

$$E(X) = \sum XP(X)$$

$$= 0 \times \frac{9995}{10000} + \frac{1500}{10000} + \frac{2000}{10000} + \frac{3000}{10000}$$

$$= \frac{1500 + 2000 + 3000}{10000}$$

$$= \frac{6500}{10000} = \frac{13}{20} = 0.65$$

**16. A bag contains 4 white and 5 black balls. Another bag contains 9 white and 7 black balls. A ball is transferred from the first bag to the second and then a ball is drawn at random from the second bag. Find the probability that the ball drawn is white.**

**Ans:** Let,  $W_1 = \{ 4 \text{ white balls} \}$

$B_1 = \{ 5 \text{ black balls} \}$

And  $W_2 = \{ 9 \text{ white balls} \}$ ,  $B_2 = \{ 7 \text{ black balls} \}$

Let  $E_1 =$  event that ball transferred from the first bag is white

and  $E_2 =$  event that the ball transferred from the first bag is black.

and  $E =$  event that the ball drawn from the second bag is white.

$$P\left(\frac{E}{E_1}\right) = \frac{10}{17} \cdot P\left(\frac{E}{E_2}\right) = \frac{9}{17}$$

$$P(E_1) = \frac{4}{9}$$

$$\text{and } P(E_2) = \frac{5}{9}$$

$$P(E) = P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right)$$

$$= \frac{4}{9} \cdot \frac{10}{17} + \frac{5}{9} \cdot \frac{9}{17} = \frac{40 + 45}{153} = \frac{85}{153} = \frac{5}{9}$$

**17. Bag I contains 3 black and 2 white balls, Bag II contains 2 black and 4 white balls. A bag and a ball is selected at random. Determine the probability of selecting a black ball.**

**Ans:** Given, Bag I = {3B, 2W}

and Bag II = {2B, 4W}

Let  $E_1$  = event that bag I is selected

$E_2$  = event that bag II is selected

And  $E$  = event that a black ball is selected

$$P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2}, P\left(\frac{E}{E_1}\right) = \frac{3}{5}, P\left(\frac{E}{E_2}\right) = \frac{2}{6} = \frac{1}{3}$$

$$\therefore P(E) = P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right)$$

$$= \frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{2}{6} = \frac{3}{10} + \frac{2}{12}$$

$$= \frac{18+10}{60} = \frac{28}{60} = \frac{7}{15}$$

**18. A box has 5 blue and 4 red balls. one ball is drawn at random and not replaced. Its colour is also not noted. Then another ball is drawn at random. What is the probability of the second ball being blue?**

**Ans:** Given, A box = (5 blue, 4 red)

Let  $E_1$  = event that first ball drawn is blue

$E_2$  = event that first ball drawn is red

and  $E$  = event that the second ball drawn is blue.

$$\therefore P(E) = P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) = \frac{5}{9} \cdot \frac{4}{8} + \frac{4}{9} \cdot \frac{5}{8} = \frac{20}{72} + \frac{20}{72} = \frac{40}{72} = \frac{5}{9}$$

**19. Four cards are successively drawn without replacement from a deck of 52 playing cards. What is the probability that all the four cards are king?**

**Ans:** Let's assume  $E_1, E_2, E_3$  and  $E_4$  be the events respectively the first, second, third and fourth card is king .

$$\begin{aligned}\therefore P(E_1 \cap E_2 \cap E_3 \cap E_4) &= P(E_1) \cdot P\left(\frac{E_2}{E_1}\right) \cdot P\left(\frac{E_3}{E_1 \cap E_2}\right) \cdot P[E_4(E_1 \cap E_2 \cap E_3) \cap E_4] \\ &= \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} \cdot \frac{1}{49} = \frac{24}{52 \cdot 51 \cdot 50 \cdot 49} \\ &= \frac{1}{13 \cdot 17 \cdot 25 \cdot 49} = \frac{1}{270725}\end{aligned}$$

**20. A die is thrown 5 times. Find the probability that an odd number will come up exactly three times.**

**Ans:** Here,  $p = \left(\frac{1}{6} + \frac{1}{6} + \frac{1}{6}\right) = \frac{1}{2}$  and  $q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}, n = 5, r = 3$

$$\therefore P(X = r) = {}^n C_r (p)^r (q)^{n-r} = {}^5 C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3} = \frac{5!}{3!2!} \cdot \frac{1}{8} \cdot \frac{1}{4} = \frac{10}{32} = \frac{5}{16}$$

**21. Ten coins are tossed. What is the probability of getting at least 8 heads?**

**Ans:** Here,  $n = 10, p = \frac{1}{2}, q = \frac{1}{2}$

$$\therefore P(X = r) = P(r = 8) + P(r = 9) + P(r = 10)$$

$$= {}^{10} C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8} + {}^{10} C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + {}^{10} C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10}$$

$$\begin{aligned}
 &= \frac{10!}{8!2!} \left(\frac{1}{2}\right)^{10} + \frac{10!}{9!1!} \left(\frac{1}{2}\right)^{10} + \frac{10!}{0!10!} \left(\frac{1}{2}\right)^{10} \\
 &= \left(\frac{1}{2}\right)^{10} \left[ \frac{10 \times 9}{2} + 10 + 1 \right] \\
 &= \left(\frac{1}{2}\right)^{10} \cdot 56 = \frac{1}{2^7 \cdot 2^3} \cdot 56 = \frac{7}{128}
 \end{aligned}$$

**22. The probability of a man hitting a target is 0.25. He shoots 7 times. What is the probability of his hitting at least twice?**

**Ans:** Here,  $n = 7$ ,  $p = 0.25 = \frac{1}{4}$ ,  $q = 1 - \frac{1}{4} = \frac{3}{4}$

$$\begin{aligned}
 \therefore P(X \geq 2) &= 1 - [P(r=0) + P(r=1)] \\
 &= 1 - \left[ {}^7C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{7-0} + {}^7C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{7-1} \right] \\
 &= 1 - \left[ \frac{7!}{0!7!} \left(\frac{3}{4}\right)^7 + \frac{7!}{1!6!} \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^6 \right] \\
 &= 1 - \left[ \left(\frac{3}{4}\right)^6 \left(\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 7\right) \right] \\
 &= 1 - \left[ \frac{3^5}{4^5} \left(\frac{10}{4}\right) \right] = 1 - \left[ \frac{3^5 \times 10}{4^7} \right] = 1 - \left[ \frac{27 \cdot 27 \cdot 10}{64 \cdot 256} \right] \\
 &= 1 - \left[ \frac{7290}{16384} \right] = 1 - \frac{3645}{8192} = \frac{4547}{8192}
 \end{aligned}$$

**23. A lot of 100 watches is known to have 10 defective watches. If 8 watches are selected (one by one with replacement) at random, what is the probability that there will be at least one defective watch?**

**Ans:** Probability (defective watch out of 100 watches) =  $\frac{10}{100} = \frac{1}{10}$

Here,  $p = \frac{1}{10}$ ,  $q = \frac{9}{10}$ ,  $n = 8$  and  $r \geq 1$

$$\begin{aligned}\therefore P(X \geq 1) &= 1 - P(x = 0) = 1 - {}^8C_0 \left(\frac{1}{10}\right)^0 \left(\frac{9}{10}\right)^{8-0} \\ &= 1 - \frac{8!}{0!8!} \left(\frac{9}{10}\right)^8 = 1 - \left(\frac{9}{10}\right)^8\end{aligned}$$

**24. Consider the probability distribution of a random variable X:**

|        |     |      |     |     |      |
|--------|-----|------|-----|-----|------|
| $X$    | 0   | 1    | 2   | 3   | 4    |
| $P(X)$ | 0.1 | 0.25 | 0.3 | 0.2 | 0.15 |

**Calculate (i)  $V\left(\frac{X}{2}\right)$**

**Ans:** We have

|           |     |      |     |     |      |
|-----------|-----|------|-----|-----|------|
| $X$       | 0   | 1    | 2   | 3   | 4    |
| $P(X)$    | 0.1 | 0.25 | 0.3 | 0.2 | 0.15 |
| $XP(X)$   | 0   | 0.25 | 0.6 | 0.6 | 0.60 |
| $X^2P(X)$ | 0   | 0.25 | 1.2 | 1.8 | 2.40 |

$$Var(X) = E(X^2) - [E(X)]^2$$

$$\text{here, } E(X) = \sum_{i=1}^n xP_i(x)$$

$$E(X) = 0 + 0.25 + 0.6 + 0.6 + 0.60 = 2.05$$

$$\text{and } E(X^2) = \sum_{i=1}^n x_i^2 P(x_i)$$

$$E(X^2) = 0 + 0.25 + 1.2 + 1.8 + 2.40 = 5.65$$

$$V\left(\frac{X}{2}\right) = \frac{1}{4} V(X) = \frac{1}{4} [5.65 - (2.05)^2]$$

$$\frac{1}{4} [5.65 - 4.2025] = \frac{1}{4} \times 1.4475 = 0.361875$$

**(ii) Variance of  $X$  .**

$$\text{Ans: } V(x) = 1.44475$$

**25. The probability distribution of a random variable  $X$  is given below:**

|        |     |               |               |               |
|--------|-----|---------------|---------------|---------------|
| $X$    | 0   | 1             | 2             | 3             |
| $P(X)$ | $k$ | $\frac{k}{2}$ | $\frac{k}{4}$ | $\frac{k}{8}$ |

**(i) Determine the value of  $k$ .**

$$\text{Ans: We know, } \sum_{i=1}^n P_i = 1, i = 1, 2, \dots, n \text{ and } p_i \geq 0$$

$$\Rightarrow k + \frac{k}{2} + \frac{k}{4} + \frac{k}{8} = 1$$

$$\Rightarrow 8k + 4k + 2k + k = 8$$

$$\therefore k = \frac{8}{15}$$

**(ii) Determine  $P(X \leq 2)$  and  $P(X > 2)$**

**Ans:**  $P(X \leq 2) = P(0) + P(1) + P(2) = k + \frac{k}{2} + \frac{k}{4}$

$$= \frac{(4k + 2k + k)}{4} = \frac{7k}{4} = \frac{7}{4} \cdot \frac{8}{15} = \frac{14}{15}$$

And  $P(X > 2) = P(3) = \frac{k}{8} = \frac{1}{8} \cdot \frac{8}{15} = \frac{1}{15}$

**(iii) Find  $P(X \leq 2) + P(X > 2)$ .**

**Ans:**  $P(X \leq 2) + P(X > 2) = \frac{14}{15} + \frac{1}{15} = 1$

**26. For the following probability distribution, determine standard deviation of the random variable  $X$ .**

|        |     |     |     |
|--------|-----|-----|-----|
| $X$    | 2   | 3   | 4   |
| $P(X)$ | 0.2 | 0.5 | 0.3 |

**Ans:**

|           |     |     |     |
|-----------|-----|-----|-----|
| $X$       | 2   | 3   | 4   |
| $P(X)$    | 0.2 | 0.5 | 0.3 |
| $XP(X)$   | 0.4 | 1.5 | 1.2 |
| $X^2P(X)$ | 0.8 | 4.5 | 4.8 |

We know that, standard deviation of  $X = \sqrt{\text{Var } X}$

where,  $\text{Var } X = E(X^2) - [E(X)]^2$

$$= \sum_{i=1}^n x_i^2 P(x_i) - \left[ \sum_{i=1}^n x_i P_i \right]^2$$

$$\therefore \text{Var } X = [0.8 + 4.5 + 4.8] - [0.4 + 1.5 + 1.2]^2$$

$$= 10.1 - (3.1)^2 = 10.1 - 9.61 = 0.49$$

$\therefore$  Standard deviation of  $X = \sqrt{\text{Var } X} = \sqrt{0.49} = 0.7$

**27. A biased die is such that  $P(4) = \frac{1}{10}$  and other scores being equally likely.**

**The die is tossed twice. If  $X$  is the 'number of four seen', find the variance of the random variable  $x$ .**

**Ans:** Here,  $X = 0, 1, 2$

$$P_{(f)} = \frac{1}{10} \text{ and } P_{(ret)} = \frac{9}{10}$$

$$\text{So, } P(X = 0) = P_{(ret)} \cdot P_{(ret)} = \frac{9}{10} \cdot \frac{9}{10} = \frac{81}{100}$$

$$P(X = 1) = P_{(ret)} \cdot P_{(f)} + P_{(f)} \cdot P_{(ret)} = \frac{9}{10} \cdot \frac{1}{10} + \frac{1}{10} \cdot \frac{9}{10} = \frac{18}{100}$$

$$P(X = 2) = P_{(f)} \cdot P_{(f)} = \frac{1}{10} \cdot \frac{1}{10} = \frac{1}{100}$$

|           |                  |                  |                 |
|-----------|------------------|------------------|-----------------|
| $X$       | 0                | 1                | 2               |
| $P(X)$    | $\frac{81}{100}$ | $\frac{18}{100}$ | $\frac{1}{100}$ |
| $XP(X)$   | 0                | $\frac{18}{100}$ | $\frac{2}{100}$ |
| $X^2P(X)$ | 0                | $\frac{1}{100}$  | $\frac{4}{100}$ |

$$\therefore \text{Var}(X) = E(X^2) - [E(X)]^2 = \sum X^2P(X) - [\sum XP(X)]^2$$

$$= \left[ 0 + \frac{18}{100} + \frac{4}{100} \right] - \left[ 0 + \frac{18}{100} + \frac{2}{100} \right]^2$$

$$= \frac{22}{100} - \left( \frac{20}{100} \right)^2 = \frac{11}{50} - \frac{1}{25}$$

$$= \frac{11-2}{50} = \frac{9}{50} = \frac{18}{100} = 0.18$$

**28. A die is thrown three times. Let X be 'the number of two seen'. Find the expectation of x.**

**Ans:** We have,  $X = 0, 1, 2, 3$ .

Ans: Here, we have  $X = 0, 1, 2, 3$

[ $\because$  die is thrown 3 times ]

and  $p = \frac{1}{6}, q = \frac{5}{6}$

$$\therefore P(X = 0) = P(\text{not } 2) \cdot P(\text{not } 2) \cdot P(\text{not } 2) = \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} = \frac{125}{216}$$

$$P(X = 1) = P(2) \cdot P(\text{not } 2) \cdot P(\text{not } 2) + P(\text{not } 2) \cdot P(2) \cdot P(\text{not } 2) + P(\text{not } 2) \cdot P(\text{not } 2) \cdot P(2)$$

$$= \frac{1}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} + \frac{5}{6} \cdot \frac{1}{6} \cdot \frac{5}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} = \frac{25}{216} + \frac{25}{216} + \frac{25}{216} = \frac{75}{216}$$

$$P(X = 2) = P(2) \cdot P(2) \cdot P(\text{not } 2) + P(2) \cdot P(\text{not } 2) \cdot P(2) + P(\text{not } 2) \cdot P(\text{not } 2) \cdot P(2)$$

$$= \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{5}{6} + \frac{1}{6} \cdot \frac{5}{6} \cdot \frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{5}{216} + \frac{5}{216} + \frac{5}{216} = \frac{15}{216}$$

$$P(X = 3) = P(2) \cdot P(2) \cdot P(2) = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}$$

Now  $E(X) = \sum_{i=1}^n p_i x_i$

$$= 0 \times \frac{125}{216} + 1 \times \frac{75}{216} + 2 \times \frac{15}{216} + 3 \times \frac{1}{216}$$

$$= 0 + \frac{75}{216} + \frac{30}{216} + \frac{3}{216} = \frac{75+30+3}{216} = \frac{108}{216} = \frac{1}{2}$$

Hence, the required expectation is  $\frac{1}{2}$ .

**29. Two biased dice are thrown together. For the first die  $P(6) = \frac{1}{2}$ , the other scores being equally likely while for the second die,  $P(1) = \frac{2}{5}$  and the other scores are equally likely. Find the probability distribution of 'the number of one seen'.**

**Ans:** Given, for first die,  $P(6) = \frac{1}{2}$  and  $P(6') = \frac{1}{2}$

$$\Rightarrow P(1) + P(2) + P(3) + P(4) + P(5) = \frac{1}{2} \Rightarrow P(1) = \frac{1}{10}$$

$$P(\bar{1}) = \frac{9}{10} [\because P(1) = P(2) = P(3) = P(4) = P(5)]$$

$$\text{For second die, } P(1) = \frac{2}{5} \text{ and } P(\bar{1}) = 1 - \frac{2}{5} = \frac{3}{5}$$

Let  $X$  = Number of one's seen

$$X = 0, P(X = 0) = P(\bar{1}) \cdot P(\bar{1}) = \frac{9}{10} \cdot \frac{3}{5} = \frac{27}{50} = 0.54$$

$$P(X = 1) = P(\bar{1}) \cdot P(1) + P(1) \cdot P(\bar{1}) = \frac{9}{10} \cdot \frac{2}{5} + \frac{1}{10} \cdot \frac{3}{5}$$

$$= \frac{18}{50} + \frac{3}{50} = \frac{21}{50} = 0.42$$

$$P(X = 2) = P(1) \cdot P(1) = \frac{1}{10} \cdot \frac{2}{5} = \frac{2}{50} = 0.04$$

Hence, the required probability distribution is as below.

|     |   |   |   |
|-----|---|---|---|
| $X$ | 0 | 1 | 2 |
|-----|---|---|---|

|        |      |      |      |
|--------|------|------|------|
| $P(X)$ | 0.54 | 0.42 | 0.04 |
|--------|------|------|------|

**30. Two probability distributions of the discrete random variable  $X$  and  $Y$  are given below:**

|        |               |               |               |               |
|--------|---------------|---------------|---------------|---------------|
| $X$    | 0             | 1             | 2             | 3             |
| $P(X)$ | $\frac{1}{5}$ | $\frac{2}{5}$ | $\frac{1}{5}$ | $\frac{1}{5}$ |

|        |               |                |               |                |
|--------|---------------|----------------|---------------|----------------|
| $Y$    | 0             | 1              | 2             | 3              |
| $P(Y)$ | $\frac{1}{5}$ | $\frac{3}{10}$ | $\frac{2}{5}$ | $\frac{1}{10}$ |

**Prove that,  $E(Y^2) = 2E(X)$**

**Ans:** We have to prove that,  $E(Y^2) = 2E(X)$

$$\therefore E(X) = \sum XP(X)$$

$$= 0 \cdot \frac{1}{5} + 1 \cdot \frac{2}{5} + 2 \cdot \frac{1}{5} + 3 \cdot \frac{1}{5} = \frac{7}{5}$$

$$\Rightarrow 2E(X) = \frac{14}{5} \dots (i)$$

$$E(Y)^2 = \sum Y^2 P(Y)$$

$$= 0 \cdot \frac{1}{5} + 1 \cdot \frac{3}{10} + 4 \cdot \frac{2}{5} + 9 \cdot \frac{1}{10}$$

$$= \frac{3}{10} + \frac{8}{5} + \frac{9}{10} = \frac{28}{10} = \frac{14}{5}$$

$$\Rightarrow E(Y)^2 = \frac{14}{5} \dots (ii)$$

Hence,  $E(Y^2) = 2E(X)$ .

**31. A factory produces bulbs. The probability that any one bulb is defective is  $\frac{1}{50}$  and they are packed in boxes of 10. From the single box, find the probability that**

**(i) none of the bulbs is defective**

**Ans:** Let  $X$  is denoting that a bulb is defective.

$$\text{Given, } n = 10, p = \frac{1}{50}, q = 1 - \frac{1}{50} = \frac{49}{50}$$

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

None of the bulbs is defective, i.e.,  $r = 0$

$$P(x = 0) = {}^{10}C_0 \left(\frac{1}{50}\right)^0 \left(\frac{49}{50}\right)^{10-0} = \left(\frac{49}{50}\right)^{10}$$

**(ii) exactly two bulbs are defective**

**Ans:** If exactly two bulbs are defective

$$\therefore P(x = 2) = {}^{10}C_2 \left(\frac{1}{50}\right)^2 \left(\frac{49}{50}\right)^{10-2} = 45 \cdot \frac{(49)^8}{(50)^{10}} = 45 \times \left(\frac{1}{50}\right)^{10} \times (49)^8$$

**(iii) more than 8 bulbs work properly.**

**Ans:** More than 8 bulbs work properly, so, We can say that less than 2 bulbs are only defective

$$\begin{aligned} P(x < 2) &= P(x = 0) + P(x = 1) \\ &= {}^{10}C_0 \left(\frac{1}{50}\right)^0 \left(\frac{49}{50}\right)^{10} + {}^{10}C_1 \left(\frac{1}{50}\right)^1 \left(\frac{49}{50}\right)^9 = \left(\frac{49}{50}\right)^{10} + \frac{1}{5} \left(\frac{49}{50}\right)^9 \\ &= \left(\frac{49}{50}\right)^9 \left(\frac{49}{50} + \frac{1}{5}\right) = \left(\frac{49}{50}\right)^9 \left(\frac{59}{50}\right) = \frac{59(49)^9}{(50)^{10}} \end{aligned}$$

**32. Suppose you have two coins which appear identical in your pocket. You know that one is fair and one is a 2-headed coin. If you take one out, toss it and get a head, what is the probability that it was a fair coin?**

**Ans:** Let  $E_1$  = event that coin is fair

$E_2$  = event that one coin is 2-headed

and  $H$  = event that the tossed coin gets a head.

$$P(E_1) = \frac{1}{2}, \quad P(E_2) = \frac{1}{2}, \quad P\left(\frac{H}{E_1}\right) = \frac{1}{2}, \quad P\left(\frac{H}{E_2}\right) = 1$$

$\therefore$  Using Bayes' Theorem, we get

$$P\left(\frac{E_1}{H}\right) = \frac{P(E_1) \cdot P\left(\frac{H}{E_1}\right)}{P(E_1) \cdot P\left(\frac{H}{E_1}\right) + P(E_2) \cdot P\left(\frac{H}{E_2}\right)}$$

$$= \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

The probability of a fair coin =  $\frac{1}{3}$

**33. Suppose that 6% of the people with blood group O are left handed and 10% of those with other blood groups are left handed, 30% of the people have blood group O. If a left handed person is selected at random, what is the probability that he/she will have blood group O?**

**Ans:** Let  $E_1$  = event that a person selected is of blood group O

$E_2$  = event that the person selected is of other group and  $H$  = event that selected person is left handed

$$\therefore P(E_1) = 0.30 \text{ and } P(E_2) = 0.70$$

$$P\left(\frac{H}{E_1}\right) = 0.06 \quad P\left(\frac{H}{E_2}\right) = 0.10$$

So, from Bayes' Theorem

$$P\left(\frac{E_1}{H}\right) = \frac{P(E_1) \cdot P\left(\frac{H}{E_1}\right)}{P(E_1) \cdot P\left(\frac{H}{E_1}\right) + P(E_2) \cdot P\left(\frac{H}{E_2}\right)}$$

$$= \frac{0.30 \times 0.06}{0.30 \times 0.06 + 0.70 \times 0.10} = \frac{0.018}{0.018 + 0.070} = \frac{0.018}{0.088} = \frac{9}{44}$$

**34. Two natural numbers  $r$  and  $s$  are drawn one at a time, without replacement from the set  $S = \{1, 2, 3, \dots, n\}$ . Find  $P(r \leq p / s \leq p)$ , where  $p \in S$ .**

**Ans:** Given,  $S = \{1, 2, 3, \dots, n\}$

$$\therefore P\left(r \leq \frac{p}{s} \leq p\right) = \frac{P(P \cap S)}{P(S)} = \frac{p-1}{n} \times \frac{n}{n-1} = \frac{p-1}{n-1}$$

Hence, the required probability is  $\frac{p-1}{n-1}$ .

**35. Find the probability distribution of the maximum of the two scores obtained when a die is thrown twice. Determine also the mean of the distribution.**

**Ans:** Let  $X$  = random variable scores when a die is thrown twice.

$$X = 1, 2, 3, 4, 5, 6, S = \{(1, 1), (1, 2), (3, 3), (3, 4), (3, 5), \dots, (6, 6)\}$$

$$\text{So, } P(X = 1) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$P(X = 2) = \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{3}{36}$$

$$P(X = 3) = \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{5}{36}$$

Similarly

$$P(X = 4) = \frac{7}{36}$$

$$P(X = 5) = \frac{9}{36}$$

$$\text{and } P(X = 6) = \frac{11}{36}$$

So, the required distribution:

|        |                |                |                |                |                |                 |
|--------|----------------|----------------|----------------|----------------|----------------|-----------------|
| $X$    | 1              | 2              | 3              | 4              | 5              | 6               |
| $P(X)$ | $\frac{1}{36}$ | $\frac{3}{36}$ | $\frac{5}{36}$ | $\frac{7}{36}$ | $\frac{9}{36}$ | $\frac{11}{36}$ |

$$\text{Mean } E(X) = \sum_{i=1}^n x_i p_i$$

$$= 1 \times \frac{1}{36} + 2 \times \frac{3}{36} + 3 \times \frac{5}{36} + 4 \times \frac{7}{36} + 5 \times \frac{9}{36} + 6 \times \frac{11}{36}$$

$$= \frac{1}{36} + \frac{6}{36} + \frac{15}{36} + \frac{28}{36} + \frac{45}{36} + \frac{66}{36} = \frac{161}{36}$$

**36. The random variable  $X$  can take only the values 0, 1, 2. If  $P(X = 0) = P(X = 1) = p$  and  $E(X^2) = E(X)$ , then find the value of  $p$ .**

**Ans:** Given,  $X = 0, 1, 2$

and  $P(X)$  at  $X = 0$  and 1 is  $p$ .

Let  $P(X)$  at  $X = 2$  is  $x$

$$\Rightarrow p + p + x = 1$$

$$\Rightarrow x = 1 - 2p$$

Now we have the following distributions:

|        |     |     |          |
|--------|-----|-----|----------|
| $X$    | 0   | 1   | 2        |
| $P(X)$ | $p$ | $p$ | $1 - 2p$ |

$$\therefore E(X) = 0.p + 1.p + 2(1 - 2p) = p + 2 - 4p = 2 - 3p$$

$$\text{and } E(X^2) = 0.p + 1.p + 4(1 - 2p) = p + 4 - 8p = 4 - 7p$$

$$\text{Given that: } E(X^2) = E(X)$$

$$4 - 7p = 2 - 3p$$

$$\Rightarrow 4p = 2$$

$$\Rightarrow p = \frac{1}{2}$$

Hence, the required value of  $p = \frac{1}{2}$

**37. Find the variance of the distribution:**

|        |               |                |               |               |               |                |
|--------|---------------|----------------|---------------|---------------|---------------|----------------|
| $X$    | 0             | 1              | 2             | 3             | 4             | 5              |
| $P(X)$ | $\frac{1}{6}$ | $\frac{5}{18}$ | $\frac{2}{9}$ | $\frac{1}{6}$ | $\frac{1}{9}$ | $\frac{1}{18}$ |

$$\text{Ans: Variance (X)} = E(X^2) - [E(X)]^2$$

$$E(X) = \sum_{i=1}^n p_i x_i$$

$$= 0 \times \frac{1}{6} + 1 \times \frac{5}{18} + 2 \times \frac{2}{9} + 3 \times \frac{1}{6} + 4 \times \frac{1}{9} + 5 \times \frac{1}{18}$$

$$= 0 + \frac{5}{18} + \frac{4}{9} + \frac{3}{6} + \frac{4}{9} + \frac{5}{18} = \frac{5+8+9+8+5}{18} = \frac{35}{18}$$

$$\text{And, } E(X^2) = 0 \times \frac{1}{6} + 1 \times \frac{5}{18} + 4 \times \frac{2}{9} + 9 \times \frac{1}{6} + 16 \times \frac{1}{9} + 25 \times \frac{1}{18}$$

$$= \frac{5}{18} + \frac{8}{9} + \frac{9}{6} + \frac{16}{9} + \frac{25}{18} = \frac{5+16+27+32+25}{18} = \frac{105}{18}$$

$$\therefore \text{Var}(X) = \frac{105}{18} - \frac{35}{18} \times \frac{35}{18} = \frac{1890 - 1225}{324} = \frac{665}{324}$$

**38. A and B throw a pair of dice alternately. A wins the game if he gets a total of 6 and B wins if she gets a total of 7. If A starts the game, find the probability of winning the game by A in the third throw of the pair of dice.**

**Ans:** Let  $A_1$  = event of getting a total of 6

$$= \{(2, 4), (4, 2), (1, 5), (5, 1), (3, 3)\}$$

and  $B_1$  = event of getting a total of 7

$$= \{(2, 5), (5, 2), (1, 6), (6, 1), (3, 4), (4, 3)\}$$

Let  $P(A_1)$  is the probability, if A wins in a throw  $= \frac{5}{36}$

and  $P(B_1)$  is the probability, if B wins in a throw  $= \frac{1}{6}$

$\therefore$  The required probability of winning A in third throw

$$= P(\bar{A}_1) \cdot P(\bar{B}_1) \cdot P(A_1) = \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{5}{36} = \frac{775}{7776}$$

**39. Two dice are tossed. Find whether the following two events A and B are independent.  $A = \{(x, y) : x + y = 11\}$  and  $B = \{(x, y) : x \neq 5\}$  where  $(x, y)$  denotes a typical sample point.**

**Ans:** We have ,

$$A = \{(x, y) : x + y = 11\} \text{ and } B = \{(x, y) : x \neq 5\}$$

$$\therefore A = \{(5, 6), (6, 5)\}$$

$$B = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

$$\Rightarrow n(A) = 2, n(B) = 30 \text{ and } n(A \cap B) = 1$$

$$\therefore P(A) = \frac{2}{36} = \frac{1}{18}$$

$$\text{and } P(B) = \frac{30}{36} = \frac{5}{6}$$

$$\Rightarrow P(A) \cdot P(B) = \frac{1}{18} \cdot \frac{5}{6} = \frac{5}{108} \text{ and } P(A \cap B) = \frac{1}{36}$$

$$P(A) \cdot P(B) \neq P(A \cap B)$$

So, events  $A$  and  $B$  are not independent.

**40. An urn contains  $m$  white and  $n$  black balls. A ball is drawn at random and is put back into the urn along with  $k$  additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. Show that the probability of drawing a white ball now does not depend on  $k$ .**

**Ans:** Let  $A$  = event of having  $m$  white and  $n$  black balls

$$E_1 = \{\text{first ball drawn of white colour}\}$$

$$E_2 = \{\text{first ball drawn of black colour}\}$$

$$\text{And, } E_3 = \{\text{second ball drawn of white colour}\}$$

$$\therefore P(E_1) = \frac{m}{m+n} \text{ and } P(E_2) = \frac{n}{m+n}$$

$$P\left(\frac{E_3}{E_1}\right) = \frac{m+k}{m+n+k} \text{ and } P\left(\frac{E_3}{E_2}\right) = \frac{m}{m+n+k}$$

$$\text{Now } P(E_3) = P(E_1) \cdot P\left(\frac{E_3}{E_1}\right) + P(E_2) \cdot P\left(\frac{E_3}{E_2}\right)$$

$$= \frac{m}{m+n} \times \frac{m+k}{m+n+k} + \frac{n}{m+n} \times \frac{m}{m+n+k}$$

$$= \frac{m}{m+n+k} \left[ \frac{m+k}{m+n} + \frac{n}{m+n} \right]$$

$$= \frac{m}{m+n+k} \left[ \frac{m+n+k}{m+n} \right] = \frac{m}{m+n}$$

So, the probability of drawing a white ball does not depend upon  $k$ .

### Long Answer Type Questions

**41. Three bags contain a number of red and white balls as follows, Bag I: 3 red balls, Bag II: 2 red balls and 1 white ball and Bag III: 3 white balls. The probability that bag  $i$  will be chosen and a ball is selected from it is  $\frac{i}{6}$ , where  $i = 1, 2, 3$ .**

**What is the probability that**

**(i) a red ball will be selected**

**Ans:** Given,

Bag I=three red balls and no white ball

Bag II=two red balls and one white ball

Bag III=no red ball and three white balls

Let  $E_1, E_2$  and  $E_3$  be the events of choosing Bag I, Bag II and Bag III respectively and a ball is drawn from it.

$$P(E_1) = \frac{1}{6}, P(E_2) = \frac{2}{6} \text{ and } P(E_3) = \frac{3}{6}$$

Let  $E$  be the event that red ball is selected

$$\therefore P(E) = P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) + P(E_3) \cdot P\left(\frac{E}{E_3}\right)$$

$$= \frac{1}{6} \cdot \frac{3}{3} + \frac{2}{6} \cdot \frac{2}{3} + \frac{3}{6} \cdot 0 = \frac{3}{18} + \frac{4}{18} = \frac{7}{18}$$

**(ii) a white ball is selected?**

**Ans:** Let  $F$  be the event that ball is selected is white

$$\therefore P(F) = 1 - P(E) \quad [P(E) + P(F) = 1]$$

$$= 1 - \frac{7}{18} = \frac{11}{18}$$

**42. Refer to Exercise Q.41 above. If a white ball is selected, what is the probability that it came from**

**(i) Bag II**

**Ans:** We will use here Bayes' Theorem

$$P(E_2 / F) = \frac{P(E_2) \cdot P(F / E_2)}{P(E_1) \cdot P(F / E_1) + P(E_2) \cdot P(F / E_2) + P(E_3) \cdot P(F / E_3)}$$

$$= \frac{\frac{2}{6} \cdot \frac{1}{3}}{\frac{1}{6} \cdot 0 + \frac{2}{6} \cdot \frac{1}{3} + \frac{3}{6} \cdot 1} = \frac{\frac{2}{18}}{\frac{2}{18} + \frac{3}{6}} = \frac{2}{11}$$

**(ii) Bag III?**

$$\begin{aligned}\text{Ans: } P(E_3 / F) &= \frac{P(E_3) \cdot P(F / E_3)}{P(E_1) \cdot P(F / E_1) + P(E_2) \cdot P(F / E_2) + P(E_3) \cdot P(F / E_3)} \\ &= \frac{\frac{3}{6} \cdot 1}{\frac{1}{6} \cdot 0 + \frac{2}{6} \cdot \frac{1}{3} + \frac{3}{6} \cdot 1} = \frac{\frac{3}{6}}{\frac{2}{18} + \frac{3}{6}} = \frac{3}{6} \times \frac{18}{11} = \frac{9}{11}\end{aligned}$$

**43. A shopkeeper sells three types of flower seeds  $A_1, A_2$  and  $A_3$ . They are sold as a mixture, where the proportions are 4: 4: 2, respectively. The germination rates of the three types of seeds are 40%, 50% and 35%. Calculate the probability**

**(i) of a randomly chosen seed to germinate**

**Ans:** Given,  $A_1 : A_2 : A_3 = 4 : 4 : 2$

$$\therefore P(A_1) = \frac{4}{10}, P(A_2) = \frac{4}{10} \text{ and } P(A_3) = \frac{2}{10}$$

where  $A_1, A_2$  and  $A_3$  are the 3 types of seeds.

Let  $E$  = event that a seed germinates and  $\bar{E}$  = event that a seed does not germinate

$$\text{So, } P\left(\frac{E}{A_1}\right) = \frac{45}{100}, P\left(\frac{E}{A_2}\right) = \frac{60}{100} \text{ and}$$

$$P\left(\frac{E}{A_3}\right) = \frac{35}{100} \text{ and } P\left(\frac{\bar{E}}{A_1}\right) = \frac{55}{100}, P\left(\frac{\bar{E}}{A_2}\right) = \frac{40}{100} \text{ and } P\left(\frac{\bar{E}}{A_3}\right) = \frac{65}{100}$$

$$P(E) = P(A_1) \cdot P\left(\frac{E}{A_1}\right) + P(A_2) \cdot P\left(\frac{E}{A_2}\right) + P(A_3)$$

$$= \frac{4}{10} \cdot \frac{45}{100} + \frac{4}{10} \cdot \frac{60}{100} + \frac{2}{10} \cdot \frac{35}{100}$$

$$= \frac{180}{1000} + \frac{240}{1000} + \frac{70}{1000} = \frac{490}{1000} = 0.49$$

(ii) that it will not germinate given that the seed is of type  $A_3$

**Ans:** From sol.(i)

$$P\left(\frac{\bar{E}}{A_3}\right) = 1 - P\left(\frac{E}{A_3}\right) = 1 - \frac{35}{100} = \frac{65}{100} = 0.65$$

(iii) that it is of the type  $A_2$  given that a randomly chosen seed does not germinate.

**Ans:** Using Bayes' Theorem,

$$P\left(\frac{A_2}{\bar{E}}\right) = \frac{P(A_2) \cdot P\left(\frac{\bar{E}}{A_2}\right)}{P(A_1) \cdot P\left(\frac{\bar{E}}{A_1}\right) + P(A_2) \cdot P\left(\frac{\bar{E}}{A_2}\right) + P(A_3) \cdot P\left(\frac{\bar{E}}{A_3}\right)}$$

$$= \frac{\frac{4}{10} \cdot \frac{40}{100}}{\frac{40}{100} + \frac{4}{10} \cdot \frac{40}{100} + \frac{2}{10} \cdot \frac{65}{100}}$$

$$= \frac{\frac{160}{1000}}{\frac{220}{1000} + \frac{160}{1000} + \frac{130}{1000}} = \frac{160}{220 + 160 + 130} = \frac{160}{510} = \frac{16}{51} = 0.314$$

**44. A letter is known to have come either from TATA NAGAR or from CALCUTTA. On the envelope, just two consecutive letters TA are visible. What is the probability that the letter came from TATA NAGAR?**

**Ans:** Let  $E_1$  = event that the letter comes from TATA NAGAR and  $E_2$  = event that the letter comes from CALCUTTA Also

$E_3$  = event that on the letter, two consecutive letters TA are visible

$\therefore P(E_1) = \frac{1}{2}$  and  $P(E_2) = \frac{1}{2}$  and  $P\left(\frac{E_3}{E_1}\right) = \frac{2}{8}$  [because For TATA NAGAR, the two consecutive letters visible are TA, AT, TA, AN, NA, AG, GA, AR]

and  $P\left(\frac{E_3}{E_2}\right) = \frac{1}{7}$

[because CALCUTA, the two consecutive letters visible are CA, AL, LC, CU, UT, TT and TA]

Now using Bayes' Theorem,

$$P\left(\frac{E_1}{E_3}\right) = \frac{P(E_1) \cdot P\left(\frac{E_3}{E_1}\right)}{P(E_1) \cdot P\left(\frac{E_3}{E_1}\right) + P(E_2) \cdot P\left(\frac{E_3}{E_2}\right)}$$

$$= \frac{\frac{1}{2} \cdot \frac{2}{8}}{\frac{1}{2} \cdot \frac{2}{8} + \frac{1}{2} \cdot \frac{1}{7}} = \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{14}} = \frac{\frac{1}{8}}{\frac{7+4}{56}} = \frac{7}{11}$$

**45. There are two bags, one of which contains 3 black and 4 white balls while the other contains 4 black and 3 white balls. A die is thrown. If it shows up 1 or 3, a ball is taken from the first bag but if it shows up any other number, a ball is chosen from the second bag. Find the probability of choosing a black ball.**

**Ans:** Let  $E_1$  = event of selecting Bag I

$E_2$  = event of selecting Bag II

$E_3$  = event that black ball is selected

$$\therefore P(E_1) = \frac{2}{6} = \frac{1}{3} \text{ and } P(E_2) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P\left(\frac{E_3}{E_1}\right) = \frac{3}{7} \text{ and } P\left(\frac{E_3}{E_2}\right) = \frac{4}{7}$$

$$\therefore P(E_3) = P(E_1) \cdot P\left(\frac{E_3}{E_1}\right) + P(E_2) \cdot P\left(\frac{E_3}{E_2}\right)$$

$$= \frac{1}{3} \cdot \frac{3}{7} + \frac{2}{3} \cdot \frac{4}{7} = \frac{3+8}{21} = \frac{11}{21}$$

**46. There are three urns containing 2 white and 3 black balls, 3 white and 2 black balls and 4 white and 1 black balls, respectively. There is an equal probability of each urn being chosen. A ball is drawn at random from the chosen urn and it is found to be white. Find the probability that the ball drawn was from the second urn.**

**Ans:** Let  $U_1 = \{\text{two white, three black balls}\}$

$U_2 = \{\text{three white, two black balls}\}$

$U_3 = \{\text{four white, one black balls}\}$

$$\therefore P(U_1) = P(U_2) = P(U_3) = \frac{1}{3}$$

Let  $H$  = event of drawing white ball chosen.

$$\text{So, } P\left(\frac{H}{U_1}\right) = \frac{2}{5}, P\left(\frac{H}{U_2}\right) = \frac{3}{5} \text{ and } P\left(\frac{H}{U_3}\right) = \frac{4}{5}$$

$$\begin{aligned} \therefore P\left(\frac{U_2}{H}\right) &= \frac{P(U_2) \cdot P\left(\frac{H}{U_2}\right)}{P(U_1) \cdot P\left(\frac{H}{U_1}\right) + P(U_2) \cdot P\left(\frac{H}{U_2}\right) + P(U_3) \cdot P\left(\frac{H}{U_3}\right)} \\ &= \frac{\frac{1}{3} \cdot \frac{3}{5}}{\frac{1}{3} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{3}{5} + \frac{1}{3} \cdot \frac{4}{5}} = \frac{\frac{3}{5}}{\frac{2}{5} + \frac{3}{5} + \frac{4}{5}} = \frac{3}{9} = \frac{1}{3} \end{aligned}$$

**47. By examining the chest X-ray, the probability that T B is detected when a person is actually suffering is 0.99. The probability of a healthy person diagnosed to have TB is 0.001. In a certain city, 1 in 1000 people suffer from**

**TB. A person is selected at random and is diagnosed to have TB. What is the probability that he actually has TB?**

**Ans:** Let  $E_1$  = event that person has TB

$E_2$  = event that person does not have TB

and  $H$  = event that the person is diagnosed to have TB.

$$\therefore P(E_1) = \frac{1}{1000} = 0.001, P(E_2) = 1 - \frac{1}{1000} = \frac{999}{1000} = 0.999$$

$$P\left(\frac{H}{E_1}\right) = 0.99, P\left(\frac{H}{E_2}\right) = 0.001$$

$$\begin{aligned} \therefore P\left(\frac{E_1}{H}\right) &= \frac{P(E_1) \cdot P\left(\frac{H}{E_1}\right)}{P(E_1) \cdot P\left(\frac{H}{E_1}\right) + P(E_2) \cdot P\left(\frac{H}{E_2}\right)} \\ &= \frac{0.001 \times 0.99}{0.001 \times 0.99 + 0.999 \times 0.001} = \frac{0.99}{0.99 + 0.999} \\ &= \frac{0.990}{0.990 + 0.999} = \frac{990}{1989} = \frac{110}{221} \end{aligned}$$

**48. An item is manufactured by three machines A, B and C. Out of the total number of items manufactured during a specified period, 50% are manufactured on machine A, 30% on B and 20% on C. 2% of items produced on A and 2% of items produced on B are defective and 3% of these produced on machine C are defective. All the items are stored at one godown. One item is drawn at random and is found to be defective. What is the probability that it was manufactured on machine A?**

**Ans:** Let  $E_1$  = event that the item is manufactured on machine A

$E_2$  = event that the item is manufactured on machine B

$E_3$  = event that the item is manufactured on machine C

Let  $H$  = event that the selected item is defective.

Using Bayes' Theorem,

$$P(E_1) = \frac{50}{100}, P(E_2) = \frac{30}{100}, P(E_3) = \frac{20}{100}$$

$$P\left(\frac{H}{E_1}\right) = \frac{2}{100}, P\left(\frac{H}{E_2}\right) = \frac{2}{100} \text{ and } P\left(\frac{H}{E_3}\right) = \frac{3}{100}$$

$$\begin{aligned} \therefore P\left(\frac{E_1}{H}\right) &= \frac{P(E_1) \cdot P\left(\frac{H}{E_1}\right)}{P(E_1) \cdot P\left(\frac{H}{E_1}\right) + P(E_2) \cdot P\left(\frac{H}{E_2}\right) + P(E_3) \cdot P\left(\frac{H}{E_3}\right)} \\ &= \frac{\frac{50}{100} \times \frac{2}{100}}{\frac{50}{100} \times \frac{2}{100} + \frac{30}{100} \times \frac{2}{100} + \frac{20}{100} \times \frac{3}{100}} \\ &= \frac{100}{100 + 60 + 60} = \frac{100}{220} = \frac{10}{22} = \frac{5}{11} \end{aligned}$$

**49. Let  $X$  be a discrete random variable whose probability distribution is defined as follows**

$$P(X = x) = \begin{cases} k(x+1) & \text{for } x = 1, 2, 3, 4 \\ 2kx & \text{for } x = 5, 6, 7 \\ 0 & \text{otherwise} \end{cases} \text{ where } k \text{ is a constant-Calculate:}$$

**(i) the value of  $k$**

**Ans:** Here,  $P(X = x) = k(x+1)$  for  $x = 1, 2, 3, 4$

Then,

$$P(X = 1) = k(1+1) = 2k;$$

$$P(X = 2) = k(2+1) = 3k$$

$$P(X = 3) = k(3+1) = 4k;$$

$$P(X = 4) = k(4+1) = 5k$$

Also,  $P(X = x) = 2kx$  for  $x = 5, 6, 7$

$$P(X = 5) = 2(5)k = 10k;$$

$$P(X = 6) = 2(6)k = 12k$$

$$P(X = 7) = 2(7)k = 14k$$

and for otherwise = 0 .

The probability distribution:

|        |     |     |     |     |      |      |      |           |
|--------|-----|-----|-----|-----|------|------|------|-----------|
| $X$    | 1   | 2   | 3   | 4   | 5    | 6    | 7    | otherwise |
| $P(X)$ | 2 k | 3 k | 4 k | 5 k | 10 k | 12 k | 14 k | 0         |

We know ,  $\sum_{i=1}^n P(X_i) = 1$

$$\text{So, } 2k + 3k + 4k + 5k + 10k + 12k + 14k = 1$$

$$\Rightarrow 50k = 1$$

$$\Rightarrow k = \frac{1}{50}$$

So, the value of  $k$  is  $\frac{1}{50}$

**(ii) E(X)**

**Ans:** Now the probability distribution:

|        |                |                |                |                |                 |                 |                 |
|--------|----------------|----------------|----------------|----------------|-----------------|-----------------|-----------------|
| $X$    | 1              | 2              | 3              | 4              | 5               | 6               | 7               |
| $P(X)$ | $\frac{2}{50}$ | $\frac{3}{50}$ | $\frac{4}{50}$ | $\frac{5}{50}$ | $\frac{10}{50}$ | $\frac{12}{50}$ | $\frac{14}{50}$ |

$$E(X) = 1 \times \frac{2}{50} + 2 \times \frac{3}{50} + 3 \times \frac{4}{50} + 4 \times \frac{5}{50} + 5 \times \frac{10}{50} + 6 \times \frac{12}{50} + 7 \times \frac{14}{50}$$

$$= \frac{2}{50} + \frac{6}{50} + \frac{12}{50} + \frac{20}{50} + \frac{50}{50} + \frac{72}{50} + \frac{98}{50} = \frac{260}{50} = \frac{26}{5} = 5.2$$

**(iii) Standard deviation of X.**

**Ans:** Standard deviation =  $\sqrt{\text{Variance}}$

$$\text{Variance} = E(X^2) - [E(X)]^2 \quad E(X^2) = 1 \times \frac{2}{50} + 4 \times \frac{3}{50} + 9 \times \frac{4}{50} + 16 \times \frac{5}{50} + 25 \times \frac{10}{50}$$

$$36 \times \frac{12}{50} + 49 \times \frac{14}{50}$$

$$36 \times \frac{12}{50} + 49 \times \frac{14}{50} = \frac{32}{50} + \frac{686}{50} = \frac{1498}{50}$$

$$= \frac{2}{50} + \frac{12}{50} + \frac{36}{50} + \frac{80}{50} + \frac{250}{50} + \frac{432}{50} + \frac{686}{50} = \frac{1498}{50}$$

$$\text{Then, Variance } (X) = \frac{1498}{50} - \left(\frac{26}{5}\right)^2$$

$$= \frac{1498}{50} - \frac{676}{25} = \frac{1498 - 1352}{50} = \frac{146}{50} = 2.92$$

$$\text{S.D} = \sqrt{2.92} = 1.7$$

**50. The probability distribution of a discrete random variable X is given as under:**

|      |               |               |                |                |                |                |
|------|---------------|---------------|----------------|----------------|----------------|----------------|
| X    | 1             | 2             | 4              | 2A             | 3A             | 5A             |
| P(X) | $\frac{1}{2}$ | $\frac{1}{5}$ | $\frac{3}{25}$ | $\frac{1}{10}$ | $\frac{1}{25}$ | $\frac{1}{25}$ |

**Calculate: (i) The value of A if E(X) = 2.94**

**Ans:** We know,  $E(X) = \sum_{i=1}^n P_i X_i$

$$\text{So, } E(X) = 1 \times \frac{1}{2} + 2 \times \frac{1}{5} + 4 \times \frac{3}{25} + 2A \times \frac{1}{10} + 3A \times \frac{1}{25} + 5A \times \frac{1}{25}$$

$$2.94 = \frac{1}{2} + \frac{2}{5} + \frac{12}{25} + \frac{A}{5} + \frac{3A}{25} + \frac{A}{5}$$

$$\Rightarrow 2.94 = 0.5 + 0.4 + 0.48 + \frac{13A}{25} = 1.38 + \frac{13A}{25}$$

$$\Rightarrow 2.94 - 1.38 = \frac{13A}{25}$$

$$\Rightarrow 1.56 = \frac{13A}{25}$$

$$\Rightarrow A = \frac{1.56 \times 25}{13} = 0.12 \times 25$$

$$\text{So, } A = 3$$

## (ii) Variance of $X$

**Ans:** Now the distribution:

|        |               |               |                |                |                |                |
|--------|---------------|---------------|----------------|----------------|----------------|----------------|
| $X$    | 1             | 2             | 4              | 6              | 9              | 15             |
| $P(X)$ | $\frac{1}{2}$ | $\frac{1}{5}$ | $\frac{3}{25}$ | $\frac{1}{10}$ | $\frac{1}{25}$ | $\frac{1}{25}$ |

$$E(X^2) = 1 \times \frac{1}{2} + 4 \times \frac{1}{5} + 16 \times \frac{3}{25} + 36 \times \frac{1}{10} + 81 \times \frac{1}{25} + 225 \times \frac{1}{25}$$

$$= \frac{1}{2} + \frac{4}{5} + \frac{48}{25} + \frac{36}{10} + \frac{81}{25} + \frac{225}{25}$$

$$= 0.5 + 0.8 + 1.92 + 3.6 + 3.24 + 9.00 = 19.06$$

$$\text{So, Variance } (X) = E(X^2) - [E(X)]^2$$

$$= 19.06 - (2.94)^2 = 19.06 - 8.64 = 10.42$$

51. The probability distribution of a random variable as under:

$$P(X = x) = \begin{cases} kx^2 & \text{for } x = 1, 2, 3 \\ 2kx & \text{for } x = 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

where  $k$  is a constant. Calculate:

(i)  $E(X)$

Ans:

|        |     |      |      |      |       |       |           |
|--------|-----|------|------|------|-------|-------|-----------|
| $X$    | 1   | 2    | 3    | 4    | 5     | 6     | otherwise |
| $P(X)$ | $k$ | $4k$ | $9k$ | $8k$ | $10k$ | $12k$ | 0         |

We know,  $\sum_{i=1}^n P(X_i) = 1$

$$\therefore k + 4k + 9k + 8k + 10k + 12k = 1 \Rightarrow 44k = 1 \Rightarrow k = \frac{1}{44}$$

$$E(X) = \sum_{i=1}^n P_i X_i = 1 \times k + 2 \times 4k + 3 \times 9k + 4 \times 8k + 5 \times 10k + 6 \times 12k$$

$$= k + 8k + 27k + 32k + 50k + 72k = 190k$$

$$= 190 \times \frac{1}{44} = \frac{95}{22} = 4.32 \text{ (approx.)}$$

(ii)  $E(3X^2)$

Ans: We know,  $\sum_{i=1}^n P(X_i) = 1$

$$\therefore k + 4k + 9k + 8k + 10k + 12k = 1 \Rightarrow 44k = 1 \Rightarrow k = \frac{1}{44}$$

$$E(3X^2) = 3[k + 4 \times 4k + 9 \times 9k + 16 \times 8k + 25 \times 10k + 36 \times 12k]$$

$$3[k + 16k + 81k + 128k + 250k + 432k] = 3[908k]$$

$$= 3 \times 908 \times \frac{1}{44} = \frac{2724}{44} = 61.9 \text{ (approx.)}$$

**(iii)  $P(X \geq 4)$**

**Ans:** We know,  $\sum_{i=1}^n P(X_i) = 1$

$$\therefore k + 4k + 9k + 8k + 10k + 12k = 1 \Rightarrow 44k = 1 \Rightarrow k = \frac{1}{44}$$

$$P(X \geq 4) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$= 8k + 10k + 12k = 30k$$

$$= 30 \times \frac{1}{44} = \frac{15}{22}$$

**52. A bag contains  $(2n + 1)$  coins. It is known that  $n$  of these coins have a head on both sides whereas the rest of the coins are fair. A coin is picked up at random from the bag and is tossed. If the probability that the toss results in a head is  $\frac{31}{42}$ , determine the value of  $n$ .**

**Ans:** Given,  $n$  coins are two headed coins and the remaining  $(n + 1)$  coins are fair.

Let  $E_1$  = event that unfair coin is selected

$E_2$  = event that the fair coin is selected

$E$  = event that the toss results in a head

$$\therefore P(E_1) = \frac{n}{2n+1} \text{ and } P(E_2) = \frac{n+1}{2n+1}$$

$$P\left(\frac{E}{E_1}\right) = 1 \text{ (sure event) and } P\left(\frac{E}{E_2}\right) = \frac{1}{2}$$

$$\begin{aligned}\therefore P(E) &= P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) \\ &= \frac{n}{2n+1} \cdot 1 + \frac{n+1}{2n+1} \cdot \frac{1}{2} = \frac{1}{2n+1} \left( n + \frac{n+1}{2} \right) \\ &= \frac{1}{2n+1} \left( \frac{2n+n+1}{2} \right) = \frac{3n+1}{2(2n+1)}\end{aligned}$$

But  $P(E) = \frac{31}{42}$  ( given )

$$\therefore \frac{3n+1}{2(2n+1)} = \frac{31}{42}$$

$$\Rightarrow \frac{3n+1}{2n+1} = \frac{31}{21}$$

$$\Rightarrow 63n + 21 = 62n + 31$$

$$\Rightarrow n = 10$$

**53. Two cards are drawn successively without replacement from a well shuffled deck of cards. Find the mean and standard deviation of the random variable X where X is the number of aces.**

**Ans:** Let  $X$  be the random variable ,  $X = 0, 1, 2$

and  $E$  = event of drawing an ace

and  $F$  = event of drawing non-ace.

$$\text{So, } P(E) = \frac{4}{52} \text{ and } P(\bar{E}) = \frac{48}{52}$$

$$P(X = 0) = P(\bar{E}) \cdot P(\bar{E}) = \frac{48}{52} \cdot \frac{47}{51} = \frac{188}{221}$$

$$P(X = 1) = P(E) \cdot P(\bar{E}) + P(\bar{E}) \cdot P(E) = \frac{4}{52} \times \frac{48}{51} + \frac{48}{52} \times \frac{4}{51} = \frac{32}{221}$$

$$P(X = 2) = P(E) \cdot P(E) = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$$

Distribution Table:

| $X$    | 0                 | 1                | 2               |
|--------|-------------------|------------------|-----------------|
| $P(X)$ | $\frac{188}{221}$ | $\frac{32}{221}$ | $\frac{1}{221}$ |

$$\text{Mean } E(X) = 0 \times \frac{188}{221} + 1 \times \frac{32}{221} + 2 \times \frac{1}{221} = \frac{32}{221} + \frac{2}{221} = \frac{34}{221} = \frac{2}{13}$$

$$E(X^2) = 0 \times \frac{188}{221} + 1 \times \frac{32}{221} + 4 \times \frac{1}{221} = \frac{32}{221} + \frac{4}{221} = \frac{36}{221}$$

$$\therefore \text{Variance} = E(X^2) - [E(X)]^2$$

$$= \frac{36}{221} - \left(\frac{2}{13}\right)^2 = \frac{36}{221} - \frac{4}{169} = \frac{468 - 68}{13 \times 221} = \frac{400}{2873}$$

$$\text{Standard deviation} = \sqrt{\frac{400}{2873}} = 0.377 \text{ (approx.)}$$

**54. A die is tossed twice. A 'success' is getting an even number on a toss. Find the variance of the number of successes.**

**Ans:** Let  $E$  = event of getting an even number on tossing a die.

$$\therefore P(E) = \frac{3}{6} = \frac{1}{2} \text{ and } P(\bar{E}) = 1 - \frac{1}{2} = \frac{1}{2}$$

Here  $X = 0, 1, 2$

$$P(X = 0) = P(\bar{E}) \cdot P(\bar{E}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(X = 1) = P(E) \cdot P(\bar{E}) + P(\bar{E}) \cdot P(E) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}$$

$$P(X = 2) = P(E) \cdot P(E) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

|        |               |               |               |
|--------|---------------|---------------|---------------|
| $X$    | 0             | 1             | 2             |
| $P(X)$ | $\frac{1}{4}$ | $\frac{2}{4}$ | $\frac{1}{4}$ |

$$E(X) = 0 \times \frac{1}{4} + 1 \times \frac{2}{4} + 2 \times \frac{1}{4} = \frac{2}{4} + \frac{2}{4} = 1$$

$$E(X^2) = 0 \times \frac{1}{4} + 1 \times \frac{2}{4} + 4 \times \frac{1}{4} = \frac{3}{2}$$

$$\text{Variance}(X) = E(X^2) - [E(X)]^2 = \frac{3}{2} - 1 = \frac{1}{2} = 0.5$$

**55. There are 5 cards numbered 1 to 5, one number on one card. Two cards are drawn at random without replacement. Let  $X$  denote the sum of the numbers on two cards drawn. Find the mean and variance of  $X$ .**

**Ans:** Sample space

$$S = \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2), (1, 4), (4, 1), (1, 5), (5, 1), (2, 4), (4, 2), (2, 5), (5, 2), (3, 4), (4, 3), (3, 5), (5, 3), (5, 4), (4, 5)\}$$

$$\text{So, } n(S) = 20$$

Let  $X$  be the random variable denoting the sum of the numbers on two cards drawn.

$$\therefore X = 3, 4, 5, 6, 7, 8, 9$$

$$\text{So, } P(X = 3) = \frac{2}{20}$$

$$P(X = 4) = \frac{2}{20}$$

$$P(X = 5) = \frac{4}{20}$$

$$P(X = 6) = \frac{4}{20}$$

$$P(X = 7) = \frac{4}{20}$$

$$P(X = 8) = \frac{2}{20}$$

$$P(X = 9) = \frac{2}{20}$$

$$\text{Mean, } E(X) = 3 \times \frac{2}{20} + 4 \times \frac{2}{20} + 5 \times \frac{4}{20} + 6 \times \frac{4}{20} + 7 \times \frac{4}{20} + 8 \times \frac{2}{20} + 9 \times \frac{2}{20}$$

$$= \frac{6}{20} + \frac{8}{20} + \frac{20}{20} + \frac{24}{20} + \frac{28}{20} + \frac{16}{20} + \frac{18}{20} = \frac{120}{20} = 6$$

$$E(X^2) = 9 \times \frac{2}{20} + 16 \times \frac{2}{20} + 25 \times \frac{4}{20} + 36 \times \frac{4}{20} + 49 \times \frac{4}{20} + 64 \times \frac{2}{20} + 81 \times \frac{2}{20}$$

$$= \frac{18}{20} + \frac{32}{20} + \frac{100}{20} + \frac{144}{20} + \frac{196}{20} + \frac{128}{20} + \frac{162}{20} = \frac{780}{20} = 39$$

$$\text{Variance } (X) = E(X^2) - [E(X)]^2 = 39 - (6)^2 = 39 - 36 = 3$$

### Objective Type Questions

Choose the correct answer from the given four options in each of the Exercises from Q. 56 to 82.

56. If  $P(A) = \frac{4}{5}$  and  $P(A \cap B) = \frac{7}{10}$ , then  $P\left(\frac{B}{A}\right)$  is equal to

(A)  $\frac{1}{10}$

(B)  $\frac{1}{8}$

(C)  $\frac{7}{8}$

(D)  $\frac{17}{20}$

**Ans:** Correct option is (C)

Because  $P(A) = \frac{4}{5}, P(A \cap B) = \frac{7}{10}$

$$\therefore P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{7}{10}}{\frac{4}{5}} = \frac{7}{8}$$

57. If  $P(A \cap B) = \frac{7}{10}$  and  $P(B) = \frac{17}{20}$ , then  $P\left(\frac{A}{B}\right)$  equals

(A)  $\frac{14}{17}$

(B)  $\frac{17}{20}$

(C)  $\frac{7}{8}$

(D)  $\frac{1}{8}$

**Ans:** Correct option is (A)

Because  $P(A \cap B) = \frac{7}{10}$  and  $P(B) = \frac{17}{20}$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{7}{10}}{\frac{17}{20}} = \frac{14}{17}$$

58. If  $P(A) = \frac{3}{10}$ ,  $P(B) = \frac{2}{5}$  and  $P(A \cup B) = \frac{3}{5}$ , then  $P\left(\frac{B}{A}\right) + P\left(\frac{A}{B}\right)$  equals

(A)  $\frac{1}{4}$

(B)  $\frac{1}{3}$

(C)  $\frac{5}{12}$

(D)  $\frac{7}{2}$

**Ans:** Correct option is (D)

Because,  $P(A) = \frac{3}{10}$ ,  $P(B) = \frac{2}{5}$  and  $P(A \cup B) = \frac{3}{5}$

$$\begin{aligned} P\left(\frac{B}{A}\right) + P\left(\frac{A}{B}\right) &= \frac{P(B \cap A)}{P(A)} + \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A) + P(B) - P(A \cup B)}{P(A)} + \frac{P(A) + P(B) - P(A \cup B)}{P(B)} \end{aligned}$$

$$\left[ \begin{aligned} &\because P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ &\text{i.e., } P(A \cap B) = P(A) + P(B) - P(A \cup B) \end{aligned} \right]$$

$$= \frac{\frac{3}{10} + \frac{2}{5} - \frac{3}{5}}{\frac{3}{10}} + \frac{\frac{3}{10} + \frac{2}{5} - \frac{3}{5}}{\frac{2}{5}}$$

$$= \frac{\frac{1}{10}}{\frac{3}{10}} + \frac{\frac{1}{10}}{\frac{2}{5}} = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

59. If  $P(A) = \frac{2}{5}$ ,  $P(B) = \frac{3}{10}$  and  $P(A \cap B) = \frac{1}{5}$ , then  $P(A' | B') \cdot P(B' | A')$

$P\left(\frac{A'}{B'}\right) \times P\left(\frac{B'}{A'}\right)$  is equal to

(A)  $\frac{5}{6}$

(B)  $\frac{5}{7}$

(C)  $\frac{25}{42}$

(D) 1

**Ans:** Correct option is (C)

Because,  $P(A) = \frac{2}{5}$ ,  $P(B) = \frac{3}{10}$  and  $P(A \cap B) = \frac{1}{5}$

$$P\left(\frac{A'}{B'}\right) = \frac{P(A' \cap B')}{P(B')} = \frac{1 - P(A \cup B)}{1 - P(B)}$$

$$= \frac{1 - [P(A) + P(B) - P(A \cap B)]}{1 - P(B)}$$

$$= \frac{1 - \left( \frac{2}{5} + \frac{3}{10} - \frac{1}{5} \right)}{1 - \frac{3}{10}}$$

$$= \frac{1 - \left( \frac{4+3-2}{10} \right)}{\frac{7}{10}} = \frac{1 - \frac{1}{2}}{\frac{7}{10}} = \frac{\frac{1}{2}}{\frac{7}{10}} = \frac{5}{7}$$

$$\text{And } P\left(\frac{B'}{A}\right) = \frac{P(B' \cap A)}{P(A')} = \frac{1 - P(A \cup B)}{1 - P(A)}$$

$$= \frac{1 - \frac{1}{2}}{1 - \frac{2}{5}} = \frac{\frac{1}{2}}{\frac{3}{5}} = \frac{5}{6} \left[ \because P(A \cup B) = \frac{1}{2} \right]$$

$$\therefore P\left(\frac{A'}{B}\right) \cdot P\left(\frac{B'}{A}\right) = \frac{5}{7} \cdot \frac{5}{6} = \frac{25}{42}$$

60. If A and B are two events such that  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{1}{3}$ ,  $P\left(\frac{A}{B}\right) = \frac{1}{4}$ , then  $P(A' \cap B')$  equals

(A)  $\frac{1}{12}$

(B)  $\frac{3}{4}$

(C)  $\frac{1}{4}$

(D)  $\frac{3}{16}$

**Ans:** Correct option is (C)

Because,  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{1}{3}$  and  $P\left(\frac{A}{B}\right) = \frac{1}{4} \therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

$$\Rightarrow P(A \cap B) = P\left(\frac{A}{B}\right) \cdot P(B) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$$

Now,  $P(A' \cap B') = 1 - P(A \cup B)$

$$= 1 - [P(A) + P(B) - P(A \cap B)]$$

$$= 1 - \left[ \frac{1}{2} + \frac{1}{3} - \frac{1}{12} \right] = 1 - \left[ \frac{6+4-1}{12} \right]$$

$$= 1 - \frac{9}{12} = \frac{3}{12} = \frac{1}{4}$$

**61. If  $P(A) = 0.4$ ,  $P(B) = 0.8$  and  $P(B|A) = 0.6$ , then  $P(A \cup B)$  is equal to**

**(A) 0.24**

**(B) 0.3**

**(C) 0.48**

**(D) 0.96**

**Ans:** Correct option is (D)

Because,  $P(A) = 0.4$ ,  $P(B) = 0.8$  and  $P(B|A) = 0.6$ ,

$$\therefore P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$$

$$\Rightarrow P(B \cap A) = P\left(\frac{B}{A}\right) \cdot P(A)$$

$$= 0.6 \times 0.4 = 0.24$$

$$\because P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.4 + 0.8 - 0.24$$

$$= 1.2 - 0.24 = 0.96$$

**62. If A and B are two events and  $A \neq \phi, B \neq \phi$ , then**

(A)  $P\left(\frac{A}{B}\right) = P(A) \times P(B)$

(B)  $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

(C)  $P\left(\frac{A}{B}\right) P\left(\frac{B}{A}\right) = 1$

(D)  $P\left(\frac{A}{B}\right) = \frac{P(A)}{P(B)}$

**Ans:** Correct option is (B)

Because, if  $A \neq \phi$  and  $B \neq \phi$ , then  $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

**63. A and B are events such that  $P(A) = 0.4, P(B) = 0.3$  and  $P(A \cup B) = 0.5$ , Then  $P(B' \cap A)$  equals**

(A)  $\frac{2}{3}$

(B)  $\frac{1}{2}$

(C)  $\frac{3}{10}$

(D)  $\frac{1}{5}$

**Ans:** Correct option is (D)

Because,  $P(A) = 0.4, P(B) = 0.3$  and  $P(A \cup B) = 0.5$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = 0.4 + 0.3 - 0.5 = 0.2$$

$$\therefore P(B' \cap A) = P(A) - P(A \cap B)$$

$$= 0.4 - 0.2 = 0.2 = \frac{1}{5}$$

64. If it is given that A and B are two events such that  $P(B) = \frac{3}{5}, P\left(\frac{A}{B}\right) = \frac{1}{2}$  and

$P(A \cup B) = \frac{4}{5}$ , then P(A) equals

(A)  $\frac{3}{10}$

(B)  $\frac{1}{5}$

(C)  $\frac{1}{2}$

(D)  $\frac{3}{5}$

**Ans:** Correct option is (C)

Because,  $P(B) = \frac{3}{5}, P\left(\frac{A}{B}\right) = \frac{1}{2}$  and  $P(A \cup B) = \frac{4}{5}$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow \frac{1}{2} = P(A \cap B) = \frac{3}{5}$$

$$\Rightarrow P(A \cap B) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$$

$$\text{And } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{4}{5} = P(A) + \frac{3}{5} - \frac{3}{10}$$

$$\Rightarrow \frac{4}{5} = P(A) + \frac{3}{5} - \frac{3}{10}$$

$$\Rightarrow P(A) = \frac{4}{5} - \frac{3}{5} + \frac{3}{10}$$

$$= \frac{8 - 6 + 3}{10}$$

$$= \frac{1}{2}$$

**65. In Exercise 64 above,  $P(B|A')$  is equal to**

(A)  $\frac{1}{5}$

(B)  $\frac{3}{10}$

(C)  $\frac{1}{2}$

(D)  $\frac{3}{5}$

**Ans:** Correct option is (D)

$$\text{Because, } P\left(\frac{B}{A'}\right) = \frac{P(B \cap A')}{P(A')} = \frac{P(B) - P(B \cap A)}{1 - P(A)}$$

$$= \frac{\frac{3}{5} - \frac{3}{10}}{1 - \frac{1}{2}} = \frac{\frac{6-3}{10}}{\frac{1}{2}} = \frac{6-3}{10} = \frac{3}{5}$$

66. If  $P(B) = \frac{3}{5}$ ,  $P\left(\frac{A}{B}\right) = \frac{1}{2}$  and  $P(A \cap B) = \frac{4}{5}$ , then  $P(A \cup B)' + P(A' \cup B) =$

(A)  $\frac{1}{5}$

(B)  $\frac{4}{5}$

(C)  $\frac{1}{2}$

(D) 1

**Ans:** Correct option is (D)

$$\text{Because, } P(B) = \frac{3}{5}, P\left(\frac{A}{B}\right) = \frac{1}{2}$$

$$\text{And } P(A \cup B) = \frac{4}{5}$$

$$\text{Since, } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A \cap B) = P\left(\frac{A}{B}\right) \cdot P(B)$$

$$= \frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$$

$$\text{Also, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A) = \frac{4}{5} - \frac{3}{5} + \frac{3}{10} = \frac{1}{2}$$

$$\therefore P(A \cup B)' = 1 - P(A \cup B) = 1 - \frac{4}{5} = \frac{1}{5}$$

$$\text{And } P(A \cup B) = 1 - P(A - B) = 1 - P(A \cap B)$$

$$= 1 - P(A) \cdot P(B)$$

$$= 1 - \frac{1}{2} \cdot \frac{2}{5} = \frac{4}{5}$$

$$\Rightarrow P(A \cup B') + P(A' \cup B) = \frac{1}{5} + \frac{4}{5} = \frac{5}{5} = 1$$

67. Let  $P(A) = \frac{7}{13}$ ,  $P(B) = \frac{9}{13}$  and  $P(A \cap B) = \frac{4}{13}$ . Then  $P\left(\frac{A'}{B}\right)P\left(\frac{A'}{B}\right)$  is equal to

(A)  $\frac{6}{13}$

(B)  $\frac{4}{13}$

(C)  $\frac{4}{9}$

(D)  $\frac{5}{9}$

**Ans:** Correct option is (D)

Because,  $P(A) = \frac{7}{13}$ ,  $P(B) = \frac{9}{13}$  and  $P(A \cap B) = \frac{4}{13}$

$$\therefore P\left(\frac{A'}{B}\right) = \frac{P(A' \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)}$$

$$= \frac{\frac{9}{13} - \frac{4}{13}}{\frac{9}{13}} = \frac{\frac{5}{13}}{\frac{9}{13}} = \frac{5}{9}$$

68. If A and B are such events that  $P(A) > 0$  and  $P(B) \neq 1$ , then  $P\left(\frac{A'}{B}\right)P\left(\frac{A'}{B'}\right)$  equals.

(A)  $1 - P\left(\frac{A}{B}\right)$

(B)  $1 - P(A'B)$

(C)  $\frac{1 - P(A \cup B)}{P(B)}$

(D)  $\frac{P(A')}{P(B')}$

**Ans:** Correct option is (C)

Because,  $P(A) > 0$  and  $P(B) \neq 1$

$$P\left(\frac{A'}{B}\right) = \frac{P(A' \cap B)}{P(B)} = \frac{1 - P(A \cup B)}{P(B)}$$

69. If A and B are two independent events with  $P(A) = \frac{3}{5}$  and  $P(B) = \frac{4}{9}$ , then  $P(A' \cap B)$  equals

(A)  $\frac{4}{15}$

(B)  $\frac{8}{45}$

(C)  $\frac{1}{3}$

(D)  $\frac{2}{9}$

**Ans:** Correct option is (D)

Because,  $P(A' \cap B) = 1 - P(A \cup B)$

$$= 1 - [P(A) + P(B) - P(A \cap B)]$$

$$= 1 - \left[ \frac{3}{5} + \frac{4}{9} - \frac{3}{5} \times \frac{4}{9} \right] [\because P(A \cap B) = P(A) \cdot P(B)]$$

$$= 1 - \left[ \frac{27 + 20 - 12}{45} \right] = 1 - \frac{35}{45} = \frac{10}{45} = \frac{2}{9}$$

70. If two events are independent, then

(A) they must be mutually exclusive

(B) the sum of their probabilities must be equal to 1

(C) (A) and (B) are both are correct

(D) None of the above is correct

**Ans:** Correct option is (D)

Because, for independent events  $A$  and  $B$  are

$$P(A \cap B) = P(A) \cdot P(B), P(A) \neq 0, P(B) \neq 0$$

So, they will not be mutually exclusive events.

In other words, two independent events having non-zero probabilities of occurrence cannot be mutually exclusive and conversely, two mutually exclusive events having non-zero probabilities of outcome cannot be independent.

71. Let  $A$  and  $B$  be two events such that  $P(A) = \frac{3}{8}$ ,  $P(B) = \frac{5}{8}$  and  $P(A \cup B) = \frac{3}{4}$ .

Then  $P\left(\frac{A}{B}\right) \cdot P\left(\frac{A'}{B}\right)$  is equal to

(A)  $\frac{2}{5}$

(B)  $\frac{3}{8}$

(C)  $\frac{3}{10}$

(D)  $\frac{6}{25}$

**Ans:** Correct option is (D)

Because,  $P(A) = \frac{3}{8}$ ,  $P(B) = \frac{5}{8}$  and  $P(A \cup B) = \frac{3}{4}$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = \frac{3}{8} + \frac{5}{8} - \frac{3}{4} = \frac{3+5-6}{8} = \frac{2}{8} = \frac{1}{4}$$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{5}{8}} = \frac{8}{20} = \frac{2}{5}$$

$$\text{And } P\left(\frac{A'}{B}\right) = \frac{P(A' \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)}$$

$$= \frac{\frac{5}{8} - \frac{1}{4}}{\frac{5}{8}} = \frac{\frac{5-2}{8}}{\frac{5}{8}} = \frac{3}{5}$$

$$\therefore P\left(\frac{A}{B}\right) \cdot P\left(\frac{A'}{B}\right) = \frac{2}{5} \cdot \frac{3}{5} = \frac{6}{25}$$

**72. If the events A and B are independent, then  $P(A \cap B)$  is equal to**

**(A)  $P(A) + P(B)$**

**(B)  $P(A) - P(B)$**

**(C)  $P(A) \cdot P(B)$**

**(D)  $\frac{P(A)}{P(B)}$**

**Ans:** Correct option is (C)

Because, if A and B are independent, then  $P(A \cap B) = P(A) \cdot P(B)$

**73. Two events E and F are independent. If  $P(E) = 0.3, P(E \cup F) = 0.5$ , then  $P(E|F) - P(F|E)$  equals**

**(A)  $\frac{2}{7}$**

(B)  $\frac{3}{35}$

(C)  $\frac{1}{70}$

(D)  $\frac{1}{7}$

**Ans:** Correct option is (C)

Because,  $P(E) = 0.3, P(E \cup F) = 0.5,$

Let  $P(F) = x$

$$\therefore P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$= P(E) + P(F) - P(E) \cdot P(F)$$

$$\Rightarrow 0.5 = 0.3 + x - 0.3x$$

$$\Rightarrow x = \frac{0.5 - 0.3}{0.7} = \frac{2}{7} = P(F)$$

$$\therefore P(E / F) - P(F / E) = \frac{P(E \cap F)}{P(F)} - \frac{P(F \cap E)}{P(E)}$$

$$= \frac{P(E \cap F) \cdot P(E) - P(F \cap E) \cdot P(F)}{P(E) \cdot P(F)}$$

$$= \frac{P(E \cap F)[P(E) - P(F)]}{P(E \cap F)} = P(E) - P(F)$$

$$= \frac{3}{10} - \frac{2}{7} = \frac{21 - 20}{70} = \frac{1}{70}$$

**74. A bag contains 5 red and 3 blue balls. If 3 balls are drawn at random without replacement the probability of getting exactly one red ball is**

(A)  $\frac{45}{196}$

(B)  $\frac{135}{392}$

(C)  $\frac{15}{56}$

(D)  $\frac{15}{29}$

**Ans:** Correct option is (C)

Probability of getting exactly one red (R) ball

$$= P_R \cdot P_{\bar{R}} \cdot P_R + P_R \cdot P_R \cdot P_{\bar{R}} + P_{\bar{R}} \cdot P_R \cdot P_R$$

$$= \frac{5}{8} \cdot \frac{3}{7} \cdot \frac{2}{6} + \frac{3}{8} \cdot \frac{5}{7} \cdot \frac{2}{6} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{5}{6}$$

$$= \frac{15}{4 \cdot 7 \cdot 6} + \frac{15}{4 \cdot 7 \cdot 6} + \frac{15}{4 \cdot 7 \cdot 6}$$

$$= \frac{5}{56} + \frac{5}{56} + \frac{5}{56} = \frac{15}{56}$$

**75. Refer to above Question 74. The probability that exactly two of the three balls were red, the first ball being red, is**

(A)  $\frac{1}{3}$

(B)  $\frac{4}{7}$

(C)  $\frac{15}{28}$

(D)  $\frac{5}{28}$

**Ans:** Correct option is (B)

Let  $E_1$  = Event that first ball being red

And  $E_2$  = Event that exactly two of three balls being red

$$\begin{aligned}\therefore P(E_1) &= P_R \cdot P_R \cdot P_R + P_R \cdot P_R \cdot P_{\bar{R}} + P_R \cdot P_{\bar{R}} \cdot P_R + P_R \cdot P_{\bar{R}} \cdot P_{\bar{R}} \\ &= \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} + \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} + \frac{5}{8} \cdot \frac{3}{7} \cdot \frac{4}{6} + \frac{5}{8} \cdot \frac{3}{7} \cdot \frac{2}{6} \\ &= \frac{60 + 60 + 60 + 30}{336} = \frac{210}{336}\end{aligned}$$

$$\begin{aligned}P(E_1 \cap E_2) &= P_R \cdot P_{\bar{R}} \cdot P_R + P_R \cdot P_R \cdot P_{\bar{R}} \\ &= \frac{5}{8} \cdot \frac{3}{7} \cdot \frac{4}{6} + \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} = \frac{120}{336}\end{aligned}$$

$$\therefore P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{\frac{120}{336}}{\frac{210}{336}} = \frac{4}{7}$$

**76. Three persons, A, B and C, fire at a target in turn, starting with A. Their probability of hitting the target are 0.4, 0.3 and 0.2 respectively. The probability of two hits is**

**(A) 0.024**

**(B) 0.188**

**(C) 0.336**

**(D) 0.452**

**Ans:** Correct option is (B)

Because,  $P(A) = 0.4, P(\bar{A}) = 0.6, P(B) = 0.3, P(\bar{B}) = 0.7, P(C) = 0.2$  and  $P(\bar{C}) = 0.8$

So, Probability of two hits  $= P_A \cdot P_B \cdot P_C + P_A \cdot P_B \cdot P_C + P_A \cdot P_B \cdot P_C$

$$= 0.4 \times 0.3 \times 0.8 + 0.4 \times 0.7 \times 0.2 + 0.6 \times 0.3 \times 0.2$$

$$= 0.096 + 0.056 + 0.036 = 0.188$$

**77. Assume that in a family, each child is equally likely to be a boy or a girl. A family with three children is chosen at random. The probability that the eldest child is a girl given that the family has at least one girl is**

(A)  $\frac{1}{2}$

(B)  $\frac{1}{3}$

(C)  $\frac{2}{3}$

(D)  $\frac{4}{7}$

**Ans:** Correct option is (D)

Because,

$$S = \{(B, B, B), (G, G, G), (B, G, G), (G, B, G), (G, G, B), (G, B, B), (B, G, B), (B, B, G)\}$$

$E_1$  = Event that a family has at least one girl, then

$$E_1 = \{(G, B, B), (B, G, B), (B, B, G), (G, G, B), (B, G, G), (G, B, G), (G, G, G)\}$$

$E_2$  = Event that the eldest child is a girl, then

$$E_2 = \{(G, B, B), (G, G, B), (G, B, G), (G, G, G)\}$$

$$\therefore E_1 \cap E_2 = \{(G, B, B), (G, G, B), (G, B, G), (G, G, G)\}$$

$$\therefore P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{\frac{4}{8}}{\frac{7}{8}} = \frac{4}{7}$$

78. A die is thrown and a card is selected at random from a deck of 52 playing cards. The probability of getting an even number on the die and a spade card is

(A)  $\frac{1}{2}$

(B)  $\frac{1}{4}$

(C)  $\frac{1}{8}$

(D)  $\frac{3}{4}$

**Ans:** Correct option is (C)

Because, let  $E_1$  = Event for getting an even number on the die

And  $E_2$  = Event that a spade card is selected

$$\therefore P(E_1) = \frac{3}{6} = \frac{1}{2} \text{ and } P(E_2) = \frac{13}{52} = \frac{1}{4}$$

$$\text{Then, } P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

79. A box contains 3 orange balls, 3 green balls and 2 blue balls. Three balls are drawn at random from the box without replacement. The probability of drawing 2 green balls and one blue ball is

(A)  $\frac{3}{28}$

(B)  $\frac{2}{21}$

(C)  $\frac{1}{28}$

(D)  $\frac{167}{168}$

**Ans:** Correct option is (A)

Because, probability of drawing 2 green balls and one blue ball =  $P_G \cdot P_G + P_G \cdot P_B \cdot P_G$

$$= P_G \cdot P_G \cdot P_B + P_B \cdot P_G \cdot P_G + P_G \cdot P_B \cdot P_G$$

$$= \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{2}{7} + \frac{2}{8} \cdot \frac{3}{7} \cdot \frac{2}{6} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{2}{6}$$

$$= \frac{1}{28} + \frac{1}{28} + \frac{1}{28} = \frac{3}{28}$$

**80. A flashlight has 8 batteries out of which 3 are dead. If two batteries are selected without replacement and tested, the probability that both are dead is**

(A)  $\frac{33}{56}$

(B)  $\frac{9}{64}$

(C)  $\frac{1}{14}$

(D)  $\frac{3}{28}$

**Ans:** Correct option is (D)

$$\text{Because, probability} = P_D \cdot P_D = \frac{3}{8} \cdot \frac{2}{7} = \frac{3}{28}$$

**81. Eight coins are tossed together. The probability of getting exactly 3 heads is**

(A)  $\frac{1}{256}$

(B)  $\frac{7}{32}$

(C)  $\frac{5}{32}$

(D)  $\frac{3}{32}$

**Ans:** Correct option is (B)

Because, Probability distribution  $P(X = r) = {}^n C_r (p)^r q^{n-r}$

Here,  $n = 8, r = 3, p = \frac{1}{2}$  and  $q = \frac{1}{2}$

$$\therefore \text{Required probability} = {}^8 C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{8-3} = \frac{8!}{5!3!} \left(\frac{1}{2}\right)^8 = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2} \cdot \frac{1}{16 \cdot 16} = \frac{7}{32}$$

**82. Two dice are thrown. If it is known that the sum of numbers on the dice was less than 6, the probability of getting a sum 3, is**

(A)  $\frac{1}{18}$

(B)  $\frac{5}{18}$

(C)  $\frac{1}{5}$

(D)  $\frac{2}{5}$

**Ans:** Correct option is (C)

Because, if let  $E_1$  = Event that the sum of numbers on the dice was less than 6

And  $E_2$  = Event that the sum of numbers on the dice is 3

$$\therefore E_1 = \{(1, 4), (4, 1), (2, 3), (3, 2), (2, 2), (1, 3), (3, 1), (1, 2), (2, 1), (1, 1)\}$$

$$\Rightarrow n(E_1) = 10$$

And

$$E_2 = \{(1, 2), (2, 1)\}$$

$$\Rightarrow n(E_2) = 2$$

$$\therefore \text{Required probability} = \frac{2}{10} = \frac{1}{5}$$

**83. Which one is not a requirement of a binomial distribution?**

- (A) There are 2 outcomes for each trial
- (B) There is a fixed number of trials
- (C) The outcomes must be dependent on each other
- (D) The probability of success must be the same for all the trials

**Ans:** Correct option is (C)

**84. Two cards are drawn from a well shuffled deck of 52 playing cards with replacement. The probability, that both cards are queens, is**

- (A)  $\frac{1}{13} \times \frac{1}{13}$
- (B)  $\frac{1}{13} + \frac{1}{13}$
- (C)  $\frac{1}{13} \times \frac{1}{17}$
- (D)  $\frac{1}{13} \times \frac{4}{51}$

**Ans:** Correct option is (A)

Because, probability =  $\frac{4}{52} \cdot \frac{4}{52} = \frac{1}{13} \times \frac{1}{13}$  [with replacement]

**85. The probability of guessing correctly at least 8 out of 10 answers on a true-false type examination is**

(A)  $\frac{7}{64}$

(B)  $\frac{7}{128}$

(C)  $\frac{45}{1024}$

(D)  $\frac{7}{41}$

**Ans:** Correct option is (B)

Because,  $P(X = r) = {}^n C_r (p)^r (q)^{n-r}$

$$n = 10, p = \frac{1}{2}, q = \frac{1}{2}$$

and  $r \geq 8$  i.e.,  $r = 8, 9, 10$

$$\Rightarrow P(X = r) = P(r = 8) + P(r = 9) + P(r = 10)$$

$$= {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right) + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10}$$

$$= \frac{10!}{8!2!} \left(\frac{1}{2}\right)^{10} + \frac{10!}{9!1!} \left(\frac{1}{2}\right)^{10} + \left(\frac{1}{2}\right)^{10}$$

$$= \left(\frac{1}{2}\right)^{10} \cdot [45 + 10 + 1] = \left(\frac{1}{2}\right)^{10} \cdot 56$$

$$= \frac{1}{16 \cdot 16} \cdot 56 = \frac{7}{128}$$

86. The probability that a person is not a swimmer is 0.3. The probability that out of 5 persons 4 are swimmers is

(A)  ${}^5C_4(0.7)^4(0.3)$

(B)  ${}^5C_1(0.7)(0.3)^4$

(C)  ${}^5C_4(0.7)(0.3)^4$

(D)  $(0.7)^4(0.3)$

**Ans:** Correct option is (A)

Because,  $\bar{p} = 0.3 \Rightarrow p = 0.7$  and  $q = 0.3, n = 5$  and  $r = 4$

Then probability  $= {}^5C_4(0.7)^4(0.3)$

87. The probability distribution of a discrete random variable  $X$  is given below:

|        |               |               |               |                |
|--------|---------------|---------------|---------------|----------------|
| $X$    | 2             | 3             | 4             | 5              |
| $P(X)$ | $\frac{5}{k}$ | $\frac{7}{k}$ | $\frac{9}{k}$ | $\frac{11}{k}$ |

The value of  $k$  is

(A) 8

(B) 16

(C) 32

(D) 48

**Ans:** Correct option is (C)

Because, we know,  $\sum P(X) = 1$

$$\Rightarrow \frac{5}{k} + \frac{7}{k} + \frac{9}{k} + \frac{11}{k} = 1$$

$$\Rightarrow \frac{32}{k} = 1$$

$$\therefore k = 32$$

**88. For the following probability distribution:**

|             |            |            |            |            |            |
|-------------|------------|------------|------------|------------|------------|
| <b>X</b>    | <b>-4</b>  | <b>-3</b>  | <b>-2</b>  | <b>-1</b>  | <b>0</b>   |
| <b>P(X)</b> | <b>0.1</b> | <b>0.2</b> | <b>0.3</b> | <b>0.2</b> | <b>0.2</b> |

**E(X) is equal to:**

**(A) 0**

**(B) -1**

**(C) -2**

**(D) -1.8**

**Ans:** Correct option is (D)

Because,  $E(X) = \sum XP(X)$

$$= -4 \times (0.1) + (-3 \times 0.2) + (-2 \times 0.3) + (-1 \times 0.2) + (0 \times 0.2)$$

$$= -0.4 - 0.6 - 0.6 - 0.2 = -1.8$$

**89. For the following probability distribution:**

|             |                |               |                |               |
|-------------|----------------|---------------|----------------|---------------|
| <b>X</b>    | <b>1</b>       | <b>2</b>      | <b>3</b>       | <b>4</b>      |
| <b>P(X)</b> | $\frac{1}{10}$ | $\frac{1}{5}$ | $\frac{3}{10}$ | $\frac{2}{5}$ |

**E(X<sup>2</sup>) is equal to**

- (A) 3
- (B) 5
- (C) 7
- (D) 10

**Ans:** Correct option is (D)

$$\begin{aligned}\text{Because, } E(X^2) &= \sum X^2 P(X) = 1 \cdot \frac{1}{10} + 4 \cdot \frac{1}{5} + 9 \cdot \frac{3}{10} + 16 \cdot \frac{2}{5} \\ &= \frac{1}{10} + \frac{4}{5} + \frac{27}{10} + \frac{32}{5} \\ &= \frac{1+8+27+64}{10} = 10\end{aligned}$$

**90. Suppose a random variable X follows the binomial distribution with parameters n and p, where  $0 < p < 1$ . If  $\frac{P(x=r)}{P(x=n-r)}$  is independent of n and r, then p equals**

- (A)  $\frac{1}{2}$
- (B)  $\frac{1}{3}$
- (C)  $\frac{1}{5}$
- (D)  $\frac{1}{7}$

**Ans:** Correct option is (A)

$$\text{Because, } P(X=r) = {}^n C_r (p)^r (q)^{n-r} = \frac{n!}{(n-r)!r!} (p)^r (1-p)^{n-r} [\because q=1-p] \dots (i)$$

$$P(X = 0) = (1 - p)^n$$

$$\text{and } P(X = n - r) = {}^n C_{n-r} (p)^{n-r} (q)^{n-(n-r)}$$

$$= \frac{n!}{(n-r)!r!} (p)^{n-r} (1-p)^{+r} [\because q = 1-p] [\because {}^n C_r = {}^n C_{n-r}] \dots (ii)$$

$$\frac{P(X=r)}{P(X=n-r)} = \frac{\frac{n!}{(n-r)!r!} p^r (1-p)^{n-r}}{\frac{n!}{(n-r)!r!} p^{n-r} (1-p)^{+r}} \text{ [ by Eqs. (i) and (ii)]}$$

$$= \left( \frac{1-p}{p} \right)^{n-r} \times \frac{1}{\left( \frac{1-p}{p} \right)^r}$$

This is independent of  $n$  and  $r$ , if

$$\frac{1-p}{p} = 1$$

$$\Rightarrow \frac{1}{p} = 2$$

$$\Rightarrow p = \frac{1}{2}$$

**91. In a college 30% students fail in physics, 25% fail in mathematics and 10% fail in both. One student is chosen at random. The probability that she fails in physics if she has failed in mathematics is**

(A)  $\frac{1}{10}$

(B)  $\frac{2}{5}$

(C)  $\frac{9}{20}$

(D)  $\frac{1}{3}$

**Ans:** Correct option is (B)

Because,  $P_{(Ph)} = \frac{30}{100} = \frac{3}{10}$ ,  $P_{(M)} = \frac{25}{100} = \frac{1}{4}$  and  $P_{(M \cap Ph)} = \frac{10}{100} = \frac{1}{10}$

$$\therefore P\left(\frac{Ph}{M}\right) = \frac{P(Ph \cap M)}{P(M)} = \frac{\frac{1}{10}}{\frac{1}{4}} = \frac{2}{5}$$

**92. A and B are two students. Their chances of solving a problem correctly are  $\frac{1}{3}$  and  $\frac{1}{4}$ , respectively. If the probability of their making a common error is,  $\frac{1}{20}$  and they obtain the same answer, then the probability of their answer to be correct is**

(A)  $\frac{1}{12}$

(B)  $\frac{1}{40}$

(C)  $\frac{13}{120}$

(D)  $\frac{10}{13}$

**Ans:** Correct option is (D)

If, we let  $E_1$  = Event that both A and B solve in the problem

$$\therefore P(E_1) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

Let  $E_2$  = Event that both A and B got incorrect solution of the problem

$$\therefore P(E_2) = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$$

Let  $E$  = Event that they got same answer

$$\text{Here, } P(E / E_1) = 1, P(E / E_2) = \frac{1}{20}$$

$$\begin{aligned} \therefore P(E_1 / E) &= \frac{P(E_1 \cap E)}{P(E)} = \frac{P(E_1) \cdot P(E / E_1)}{P(E_1) \cdot P(E / E_1) + P(E_2) \cdot P(E / E_2)} \\ &= \frac{\frac{1}{12} \times 1}{\frac{1}{12} \times 1 + \frac{1}{2} \times \frac{1}{20}} = \frac{\frac{1}{12}}{\frac{10+3}{120}} = \frac{120}{12 \times 3} = \frac{10}{13} \end{aligned}$$

**93. A box has 100 pens of which 10 are defective. What is the probability that out of a sample of 5 pens drawn one by one with replacement at most one is defective?**

(A)  $\left(\frac{9}{10}\right)^5$

(B)  $\frac{1}{2} \left(\frac{9}{10}\right)^4$

(C)  $\frac{1}{2} \left(\frac{9}{10}\right)^5$

(D)  $\left(\frac{9}{10}\right)^5 + \frac{1}{2} \left(\frac{9}{10}\right)^4$

**Ans:** Correct option is (D)

$$\text{Because, } n = 5, p = \frac{10}{100} = \frac{1}{10} \text{ and } q = \frac{9}{10}, r \leq 1$$

$$\Rightarrow r = 0, 1$$

Also,  $P(X = r) = {}^nC_r p^r q^{n-r}$

$$\therefore P(X = r) = P(r = 0) + P(r = 1)$$

$$= {}^5C_0 \left(\frac{1}{10}\right)^0 \left(\frac{9}{10}\right)^5 + {}^5C_1 \left(\frac{1}{10}\right)^1 \left(\frac{9}{10}\right)^4$$

$$= \left(\frac{9}{10}\right)^5 + 5 \cdot \frac{1}{10} \cdot \left(\frac{9}{10}\right)^4$$

$$= \left(\frac{9}{10}\right)^5 + \frac{1}{2} \left(\frac{9}{10}\right)^4$$

**State True or False for the statements in each of the Exercises 94 to 103.**

**94. Let  $P(A) > 0$  and  $P(B) > 0$ . Then A and B can be both mutually exclusive and independent.**

**Ans:** False

For events to be mutually exclusive -

$$P(A \cup B) = P(A) + P(B)$$

But as per the conditions in question, it is not necessary that they will meet the condition because it might be possible that

$$P(A \cap B) \neq 0$$

For events to be independent-

$$P(A \cap B) = P(A)P(B)$$

Again  $P(A) > 0$  and  $P(B) > 0$  are not sufficient conditions to validate them.

**95. If A and B are independent events, then  $A'$  and  $B'$  are also independent.**

**Ans:** True

Since if  $A$  and  $B$  are independent then,  $P(A \cap B) = P(A)P(B)$

Now we know that  $A = A \cap (B \cup B') = (A \cap B) \cup (A \cap B')$

Hence  $P(A) = P((A \cap B) \cup (A \cap B'))$

Since the events  $(A \cap B)$  and  $(A \cap B')$  are disjoint, these events are mutually exclusive

Hence  $P(A) = P((A \cap B) \cup (A \cap B')) = P(A \cap B) + P(A \cap B')$

Hence, we have

$$P(A) = P(A)P(B) + P(A \cap B')$$

$$\Rightarrow P(A \cap B') = P(A)(1 - P(B))$$

We know that  $P(B') = 1 - P(B)$

Hence we have  $P(A \cap B') = P(A)P(B')$

Hence the events  $A$  and  $B'$  are independent,

Since  $A$  and  $B'$  are independent, we have  $A'$  and  $B'$  are also independent.

**96. If  $A$  and  $B$  are mutually exclusive events, then they will be independent also.**

**Ans:** False

If  $A$  and  $B$  are mutually exclusive, then if  $A$  occurs, then  $B$  cannot occur. Then if two events are mutually exclusive, they cannot be independent. So, the correct option is B.

**97. Two independent events are always mutually exclusive.**

**Ans:** False

Two events A and B are said to be independent if the occurrence of one does not affect the probability of the probability of the occurrence of the other. Thus if A and B are two independent events, the  $P(A \cup B) = P(A).P(B)$  But two events are mutually exclusive events if the occurrence of one vent precludes the occurrence of the other event.

**98. If A and B are two independent events then  $P(A \text{ and } B) = P(A)P(B)$ .**

**Ans:** True

Two events A and B are said to be independent if the occurrence of one does not affect the probability of the probability of the occurrence of the other. Thus if A and B are two independent events, the  $P(A \cup B) = P(A).P(B)$

**99. Another name for the mean of a probability distribution is expected value.**

**Ans:** True

$$E(X) = \sum X P(X) = \mu$$

Mean gives the average of values and if it is related with probability or random variable it is often called expected value.

**100. If A and  $B'$  are independent events, then  $P(A \cup B) = 1 - P(A)P(B')$**

**Ans:** True

$$P(A' \cup B) = 1 - P(A \cap B') = 1 - P(A)P(B')$$

**101. If A and B are independent, then  $P(\text{ exactly one of A,B occurs }) = P(A)P(B') + P(B)P(A')$**

**Ans:** False

We have,  $P(\text{ exactly one of A, B occurs })$

$$= P[(A \cap B') \cup (A' \cap B)] = P(A \cap B') + P(A' \cap B)$$

$$= P(A) - P(A \cap B) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

$$= P(A \cup B) - P(A \cap B)$$

$$\text{Also, } P[\text{exactly one of A, B occurs}] = [1 - P(A' \cap B')] - [1 - P(A' \cup B')]$$

$$= P(A' \cup B') - P(A' \cap B')$$

$$= P(A') + P(B') - 2P(A' \cap B')$$

**102. If A and B are two events such that  $P(A) > 0$  and  $P(A) + P(B) > 1$ , then**

$$P\left(\frac{B}{A}\right) \geq 1 - \frac{P(B')}{P(A)}$$

**Ans:** False

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{P(A) + P(B) - P(A \cup B)}{P(A)} > 1 - \frac{P(A \cup B)}{P(A)}$$

$$\text{So, } P\left(\frac{B}{A}\right) > 1 - \frac{P(A \cup B)}{P(A)}$$

**103. If A, B and C are three independent events such that**

$$P(A) = P(B) = P(C) = p, \text{ then } P(\text{At least two of A, B, C occur}) = 3p^2 - 2p^3$$

**Ans:** True

$$P(\text{At least two of A, B, C occur})$$

$$\begin{aligned}
 &= p \times p \times (1-p) + (1-p) \cdot p \cdot p + p(1-p) \cdot p + p \cdot p \cdot p \\
 &= p^2[1-p+1-p+1-p+p] \\
 &= p^2(3-3p) + p^3 \\
 &= 3p^2 - 3p^3 + p^3 = 3p^2 - 2p^3
 \end{aligned}$$

**Fill in the blanks in each of the following questions:**

**104. If A and B are two events such that  $P(A/B) = p$ ,  $P(A) = p$ ,  $P(B) = \frac{1}{3}$  and  $P(A \cup B) = \frac{5}{9}$ , then  $p = \dots$**

**Ans:** Given,  $P(A) = p$ ,  $P(B) = \frac{1}{3}$  and  $P(A \cup B) = \frac{5}{9}$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = p$$

$$\Rightarrow P(A \cap B) = \frac{p}{3}$$

$$\text{and } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{5}{9} = p + \frac{1}{3} - \frac{p}{3}$$

$$\Rightarrow \frac{5}{9} - \frac{1}{3} = \frac{2p}{3}$$

$$\Rightarrow \frac{5-3}{9} = \frac{2p}{3}$$

$$\Rightarrow p = \frac{2}{9} \times \frac{3}{2} = \frac{1}{3}$$

**105. If A and B are such that  $P(A' \cup B') = \frac{2}{3}$  and  $P(A \cup B) = \frac{5}{9}$ , then  $P(A') + P(B') = \dots\dots$**

**Ans:** Given,  $P(A' \cup B') = \frac{2}{3}$  and  $P(A \cup B) = \frac{5}{9}$

$$P(A \cup B') = 1 - P(A \cap B)$$

$$\Rightarrow \frac{2}{3} = 1 - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\therefore P(A') + P(B) = 1 - P(A) + 1 - P(B)$$

$$= 2 - [P(A) + P(B)]$$

$$= 2 - [P(A \cup B) + P(A \cap B)]$$

$$= 2 - \left( \frac{5}{9} + \frac{1}{3} \right) = 2 - \left( \frac{5+3}{9} \right)$$

$$= \frac{18-8}{9} = \frac{10}{9}$$

**106. If X follows binomial distribution with parameters  $n = 5, p$  and  $P(X = 2) = 9 \cdot P(X = 3)$ , then  $p = \dots$**

**Ans:** because  $P(X = 2) = 9 \cdot P(X = 3)$  (where,  $n = 5$  and  $q = 1 - p$ )

$$\Rightarrow {}^5C_2 p^2 (1-p)^3 = 9 \cdot {}^5C_3 p^3 (1-p)^2$$

$$\Rightarrow \frac{5!}{2!3!} p^2 (1-p)^3 = 9 \cdot \frac{5!}{3!2!} p^3 (1-p)^2$$

$$\Rightarrow \frac{5!}{2!3!} p^2 (1-p)^3 = 9 \cdot \frac{5!}{3!2!} p^3 (1-p)^2$$

$$\Rightarrow \frac{p^2 (1-p)^3}{p^3 (1-p)^2} = 9$$

$$\Rightarrow \frac{p^2 (1-p)^3}{p^3 (1-p)^2} = 9$$

$$\Rightarrow \frac{(1-p)}{p} = 9$$

$$\Rightarrow 9p + p = 1$$

$$\Rightarrow \frac{(1-p)}{p} = 9$$

$$\Rightarrow 9p + p = 1$$

$$\therefore p = \frac{1}{10} p = \frac{1}{10}$$

**107. Let  $X$  be a random variable taking values  $x_1, x_2, \dots, x_n$  with probabilities  $P_1, P_2, \dots, P_n$ , respectively. Then  $\text{var}(X) = \dots$**

**Ans:** Because,  $\text{Var}(X) = (X^2) - [E(X)]^2$

$$= \sum_{i=1}^n X^2 P(X) - \left[ \sum_{i=1}^n X P(X) \right]^2$$

$$= \sum P_i x_i^2 - \left( \sum P_i x_i \right)^2$$

**108. Let A and B be two events. If  $P(AB) = P(A)$ , then A is..... of B.**

**Ans:**  $\because P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

$$\Rightarrow P(A) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A) \cdot P(B) = P(A \cap B)$$

So, A is independent of B.